

Geospatial Intelligence

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Xavier Reyda, Thomas Wickham, Sebastian Lemansky,
James Gotch



Data source 1: University of Maryland GTD

 start.umd.edu/data-tools/GTD

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START 
NATIONAL CONSORTIUM FOR THE
STUDY OF TERRORISM AND RESPONSES TO TERRORISM

*A Department of Homeland Security Emeritus Center
of Excellence led by the University of Maryland*

[Research](#) [Education](#) [Training](#) [Publications](#) [Data & Tools](#)

A consortium of researchers dedicated to improving the understanding of the human causes and consequences

Global Terrorism Database (GTD)

[ABOUT GTD](#) [USING GTD](#) [DOWNLOAD GTD](#) [RESEARCH](#) [FAQ](#) [TERMS OF USE](#)

ABOUT GTD

[OVERVIEW OF THE GTD](#) | [HISTORY OF THE GTD](#) | [GTD TEAM ALUMNI](#)

Overview of the GTD

The Global Terrorism Database™ (GTD) is an open-source database including information on terrorist events around the world from 1970 through 2020. Unlike many other event databases, the GTD includes systematic data on domestic as well as transnational and international terrorist incidents that have occurred during this time period and now includes more than 200,000 cases. For each GTD incident, information is available on the date and location of the incident, the weapons used and nature of the target, the number of casualties, and—when identifiable—the group or individual responsible.

Statistical information contained in the Global Terrorism Database is based on reports from a variety of open media sources. Information is not added to the GTD unless and until we have determined the sources are credible. Users should not infer any additional actions or results beyond what is presented in a GTD entry and specifically, users should not infer an individual associated with a particular incident was tried and convicted of terrorism or any other criminal offense. If new documentation about an event becomes available, an entry may be modified, as necessary and appropriate.

The National Consortium for the Study of Terrorism and Responses to Terrorism (START) makes the GTD available via this online interface in an effort to increase understanding of terrorist violence so that it can be more readily studied and defeated.

209K terror
attacks
1970-2020



Data source 2: ACLED Conflict Data



836K
conflict (Six
countries)
2016-2026

34K(Iran)
2016-2026

The Armed Conflict Location & Event Data Project (ACLED) is a registered non-profit organization that collects real-time data on the locations, dates, actors, fatalities, and types of political violence and protests worldwide. Founded in 2005, it provides a free, open-access, and widely used database, interactive mapping, and analysis tools to track global conflict patterns.



Data source 3: ACLED Conflict Data

- Iranian critical infrastructure dataset prepared by Dr. Behsat Ekici
- Iran shape files (Iran Provinces)
- CIA World Fact Book
- Other relevant data

Country Introduction



Map of Iraq





A Brief Introduction of Iraq

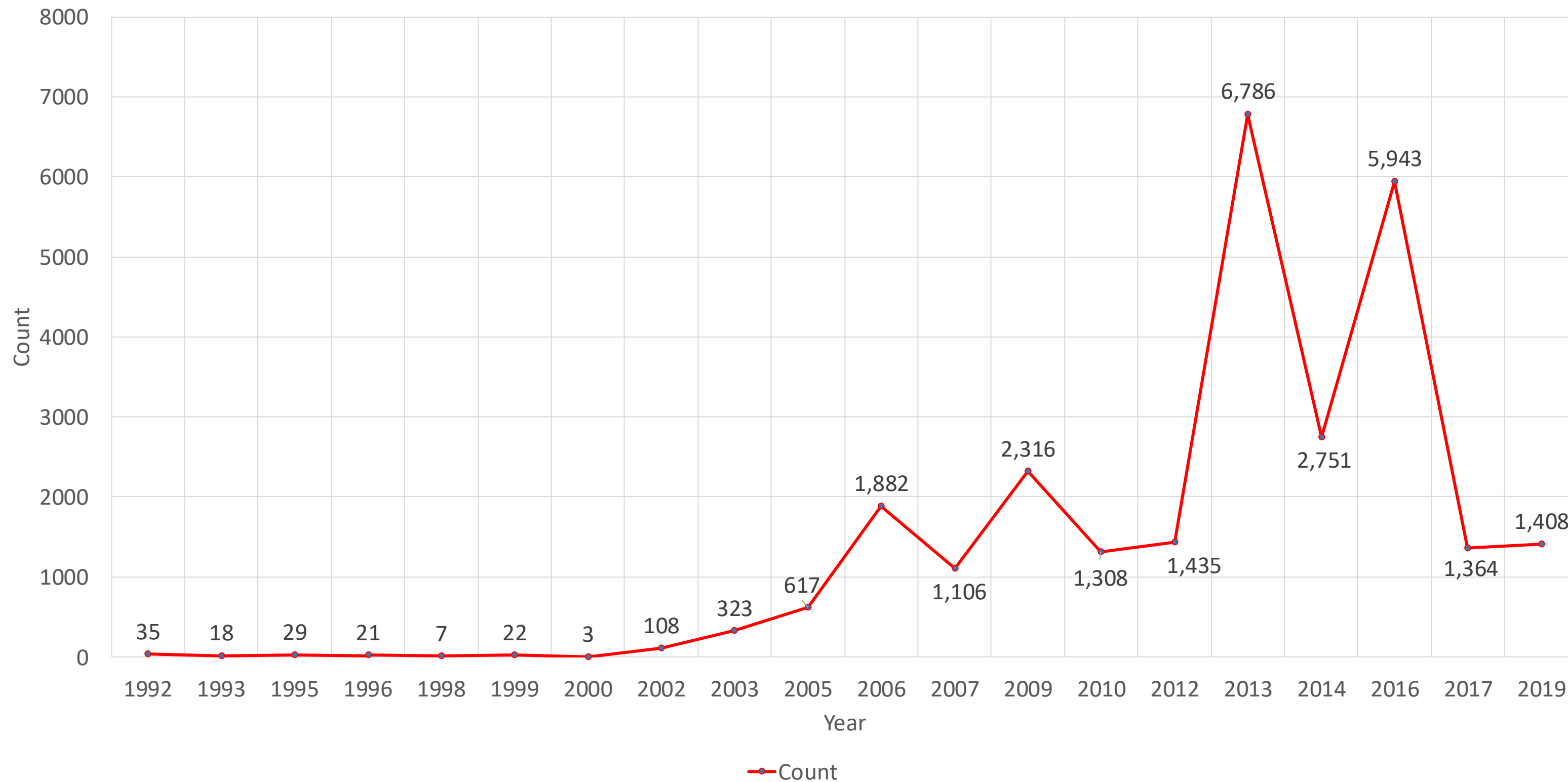
- Federal parliamentary republic
 - Power is shared between a central government and religious governments (like Kurdistan)
- Leaders
 - President – Abdul Latif Rashid
 - Prime Minister – Mohammed Shia' Al Sudani
- Ethnic Composition
 - Arab(75-80%)
 - Kurdish (15-20%)
 - other smaller groups
 - Turkmen, Assyrians, Yazdis
- Religious Composition
 - Muslim
 - Shia about 60-65%
 - Sunni about 20-35%
 - Other smaller groups
 - Christians, Yazdis
- Population
 - 45 Million people
- Economics
 - GDP Nominal
 - \$279 Billion
 - Purchasing power parity
 - GDP per PPP: about \$12,700-\$14,400
- HDI (Human Development Index)
 - Around 0.69 (medium development)
- Military Strength
 - Large military with army, air force, and police forces
 - Supported by militias and international allies
 - Has improved since fighting groups like the Islamic State of Iraq the Levent (ISIL)

Overall Distribution



How many terrorism incidents occurred in Iraq since 1970s?

Iraq Terrorism Events 1975-2020

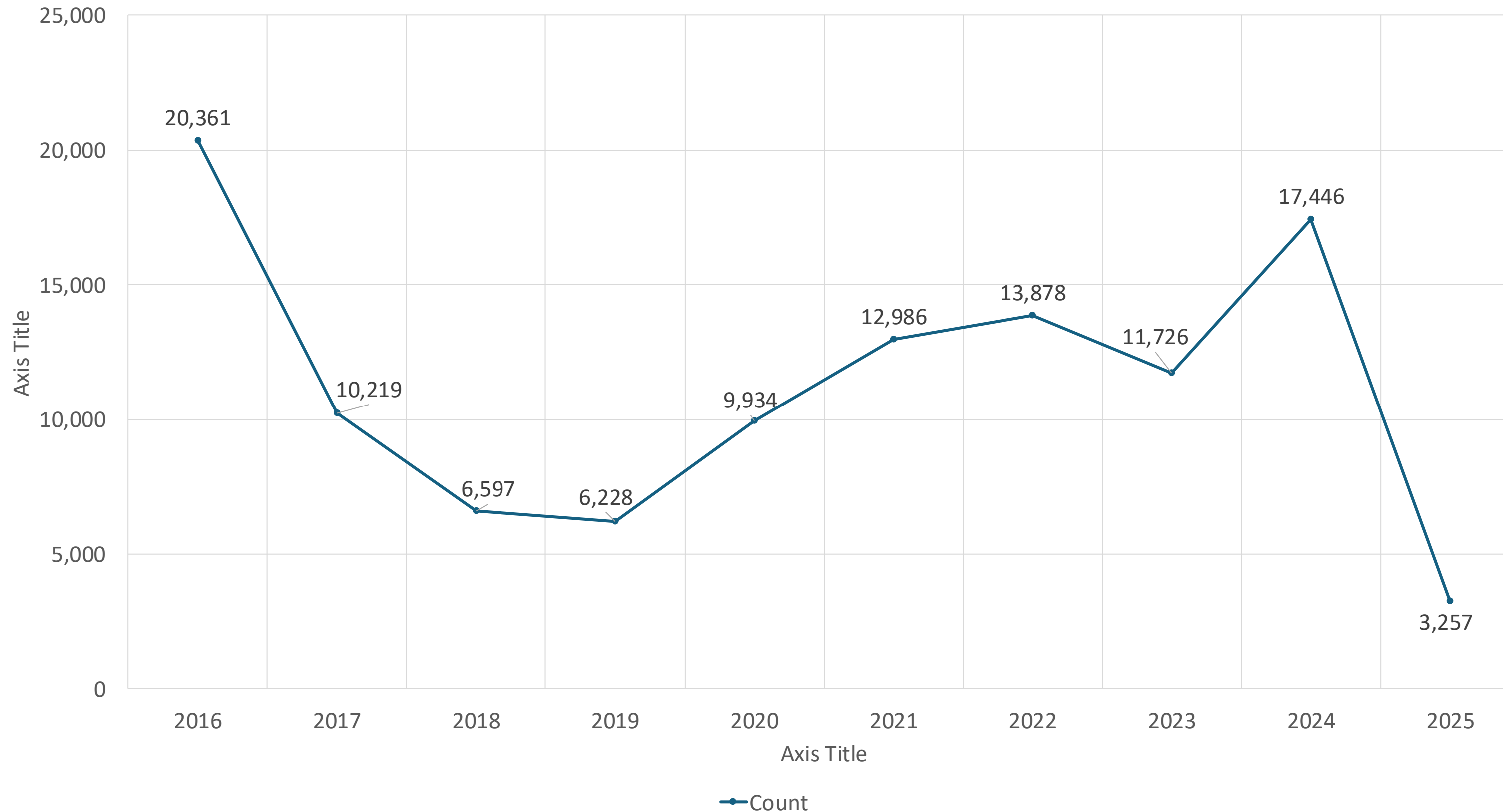


27,521



How many conflicts occurred in Iraq since 2016?

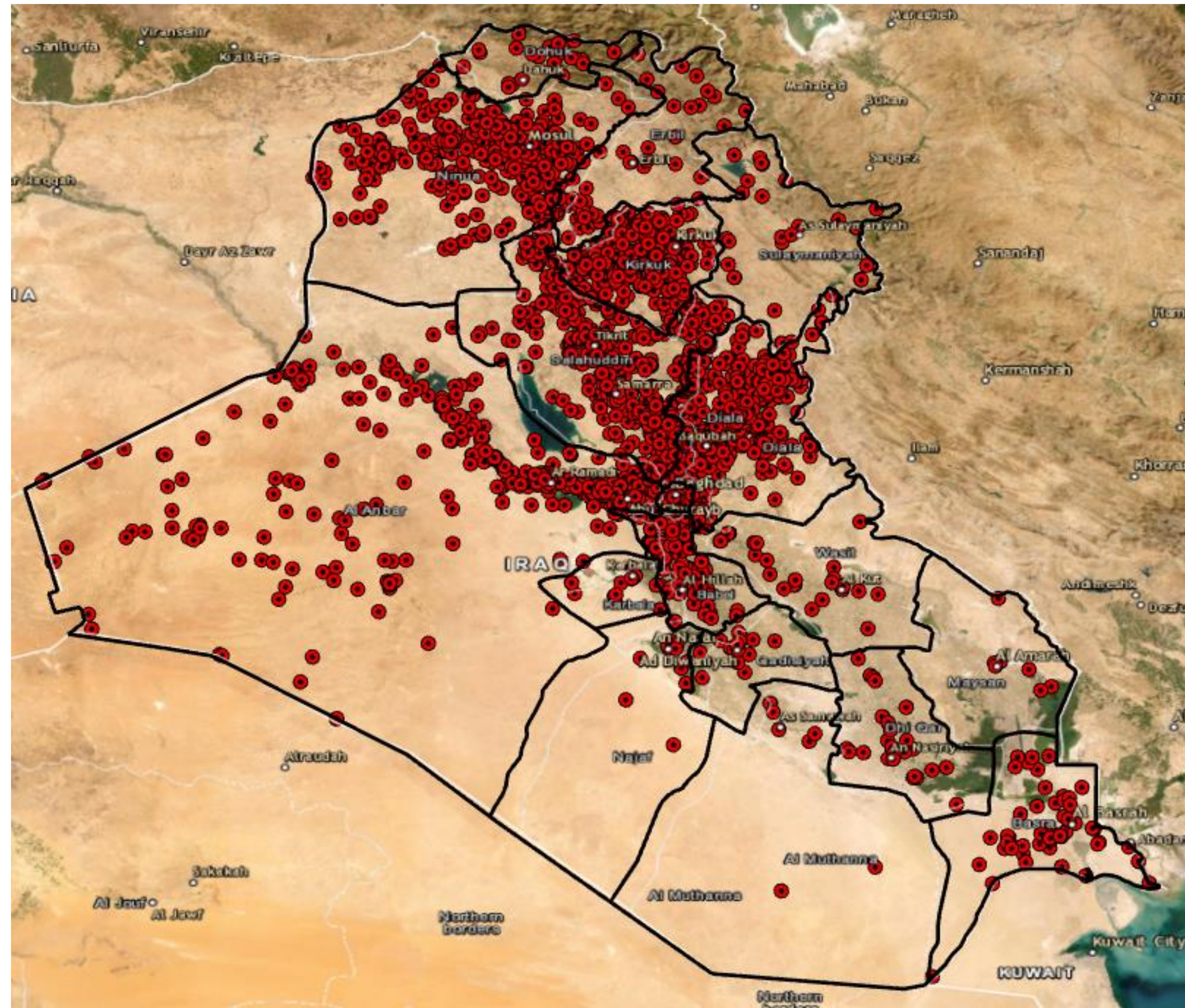
Iraq Conflicts 2016-2025



112,632

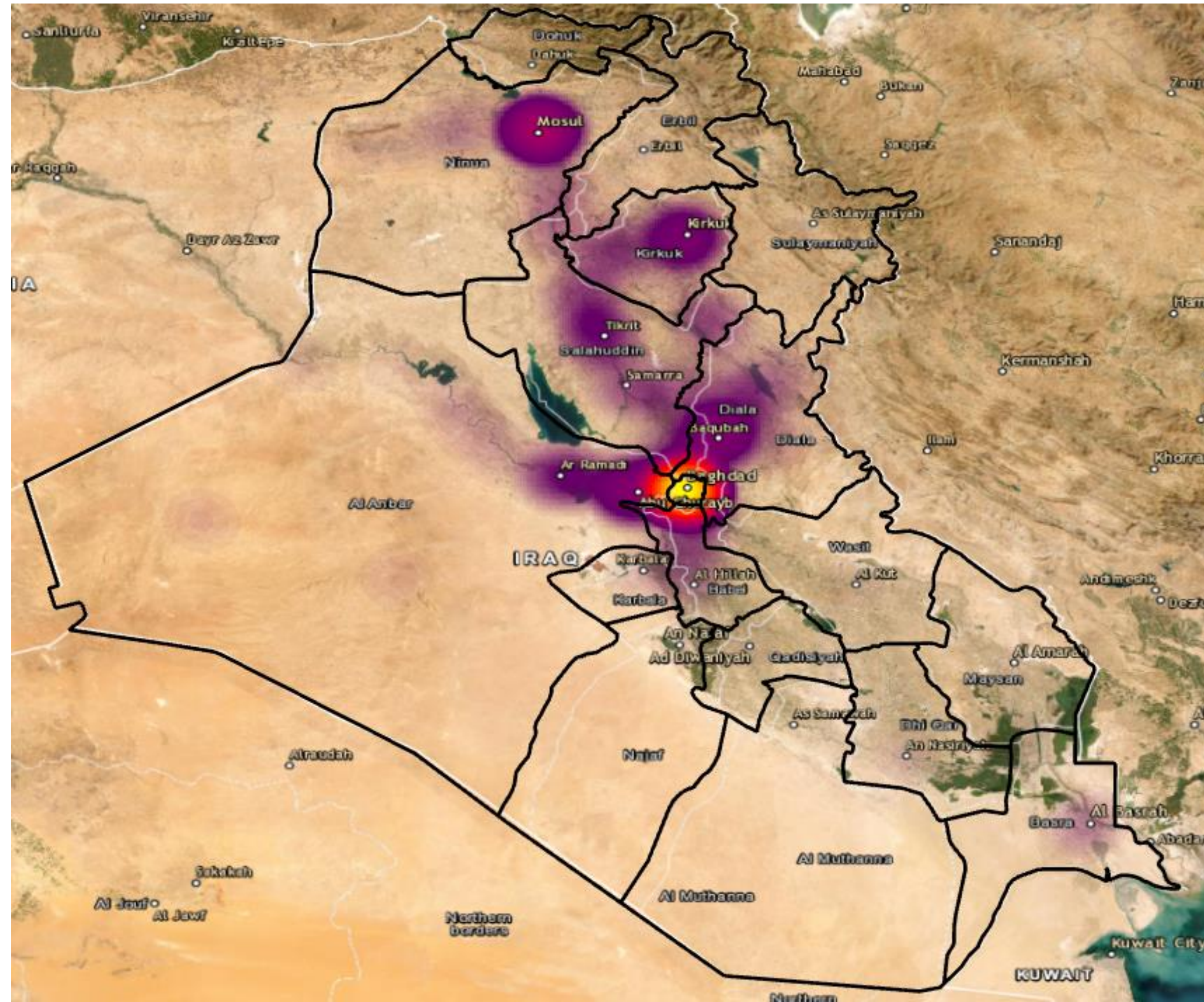


How are the terrorism incidents spread across Iraq?



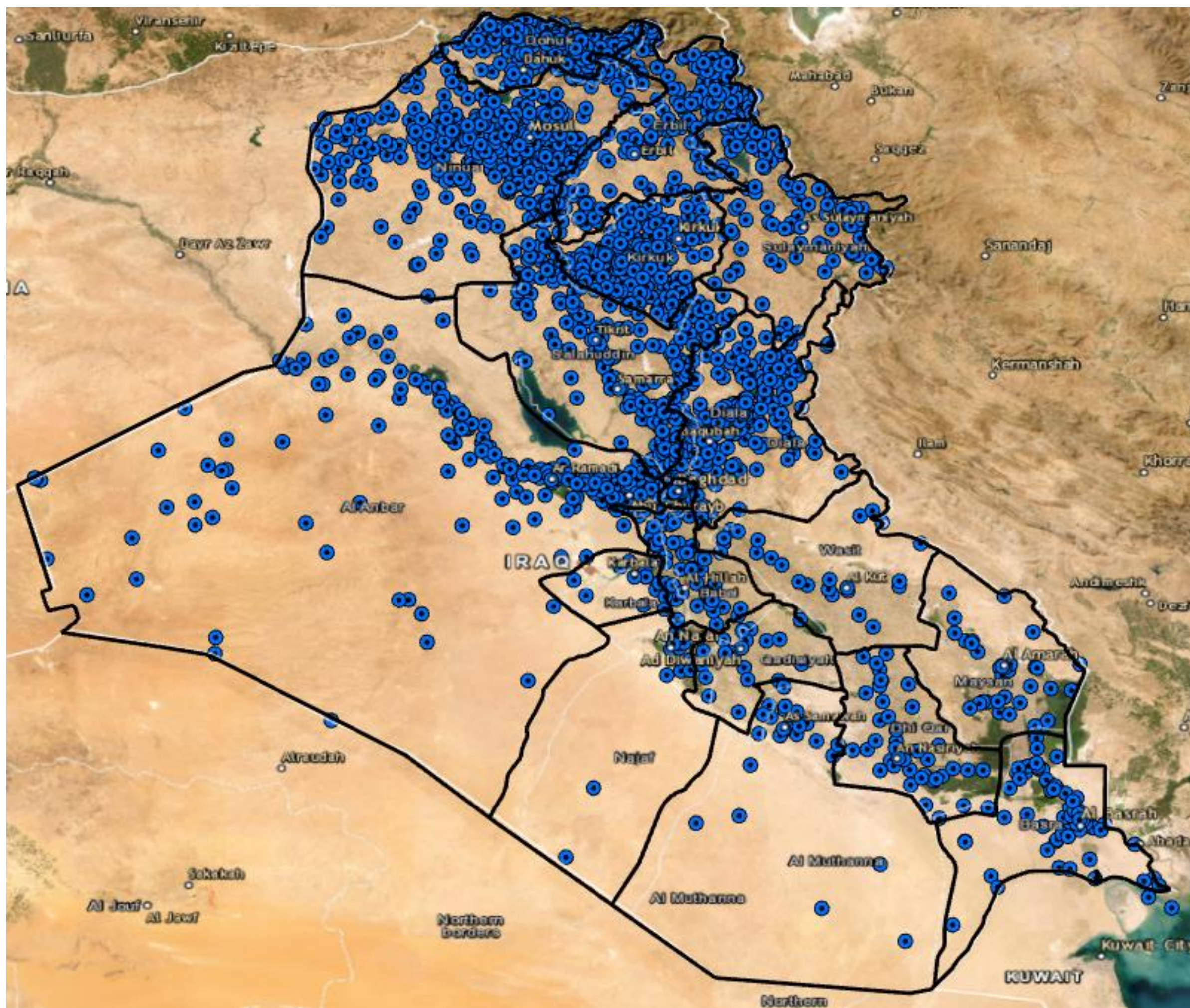


How are the terrorism incidents spread across Iraq?



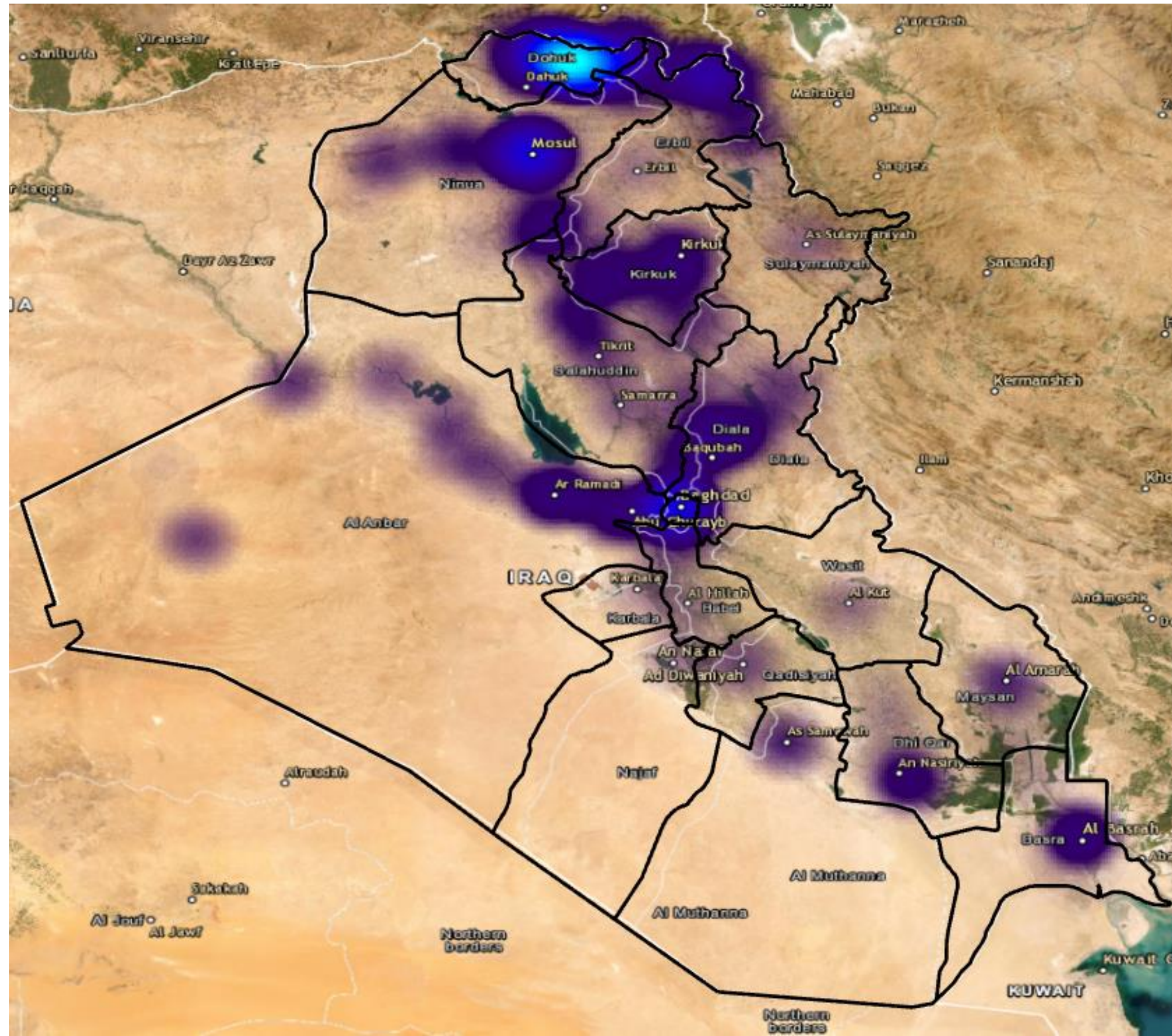


How are the conflicts spread across Iraq?

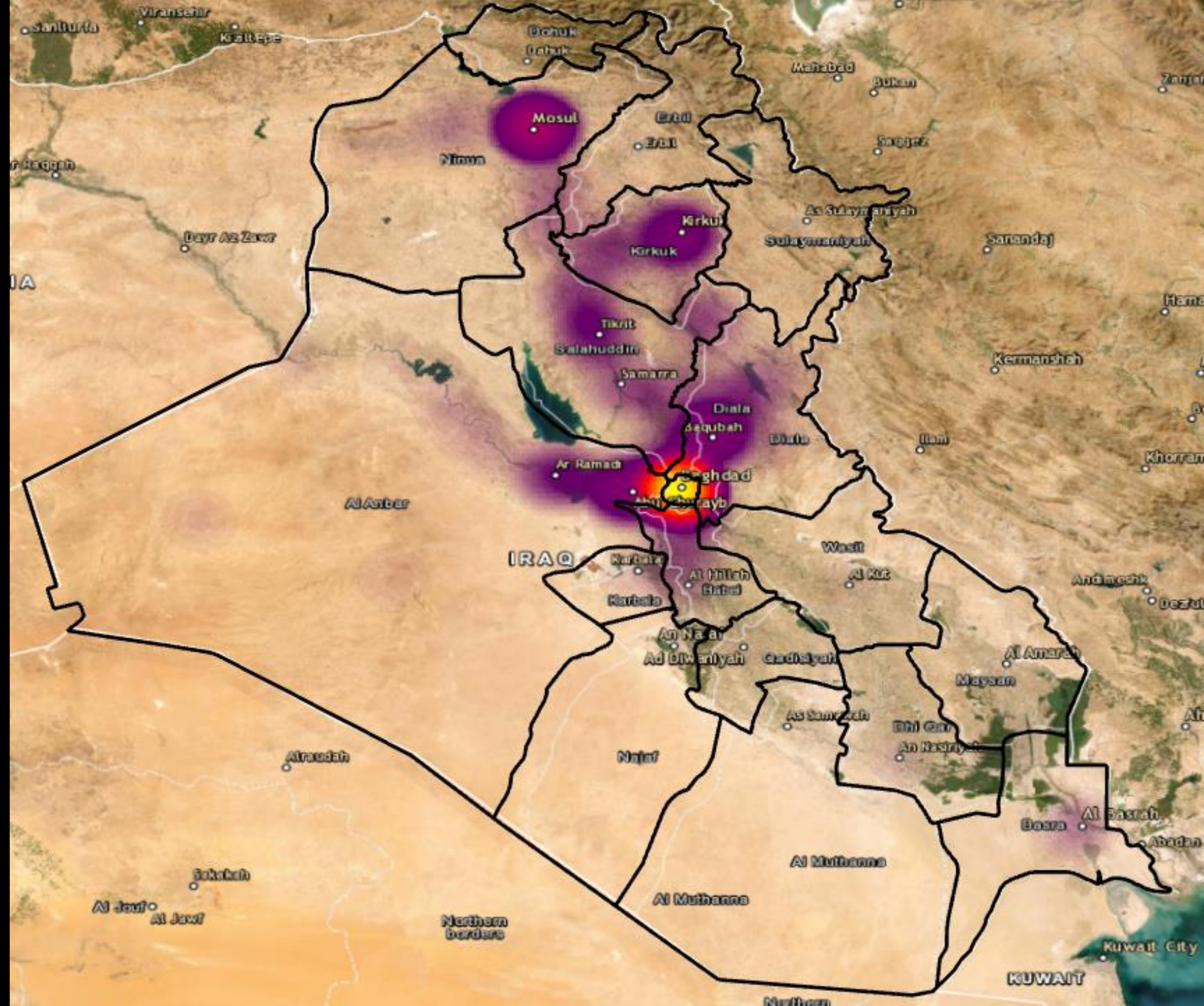




How are the conflicts spread across Iraq?



Terrorism





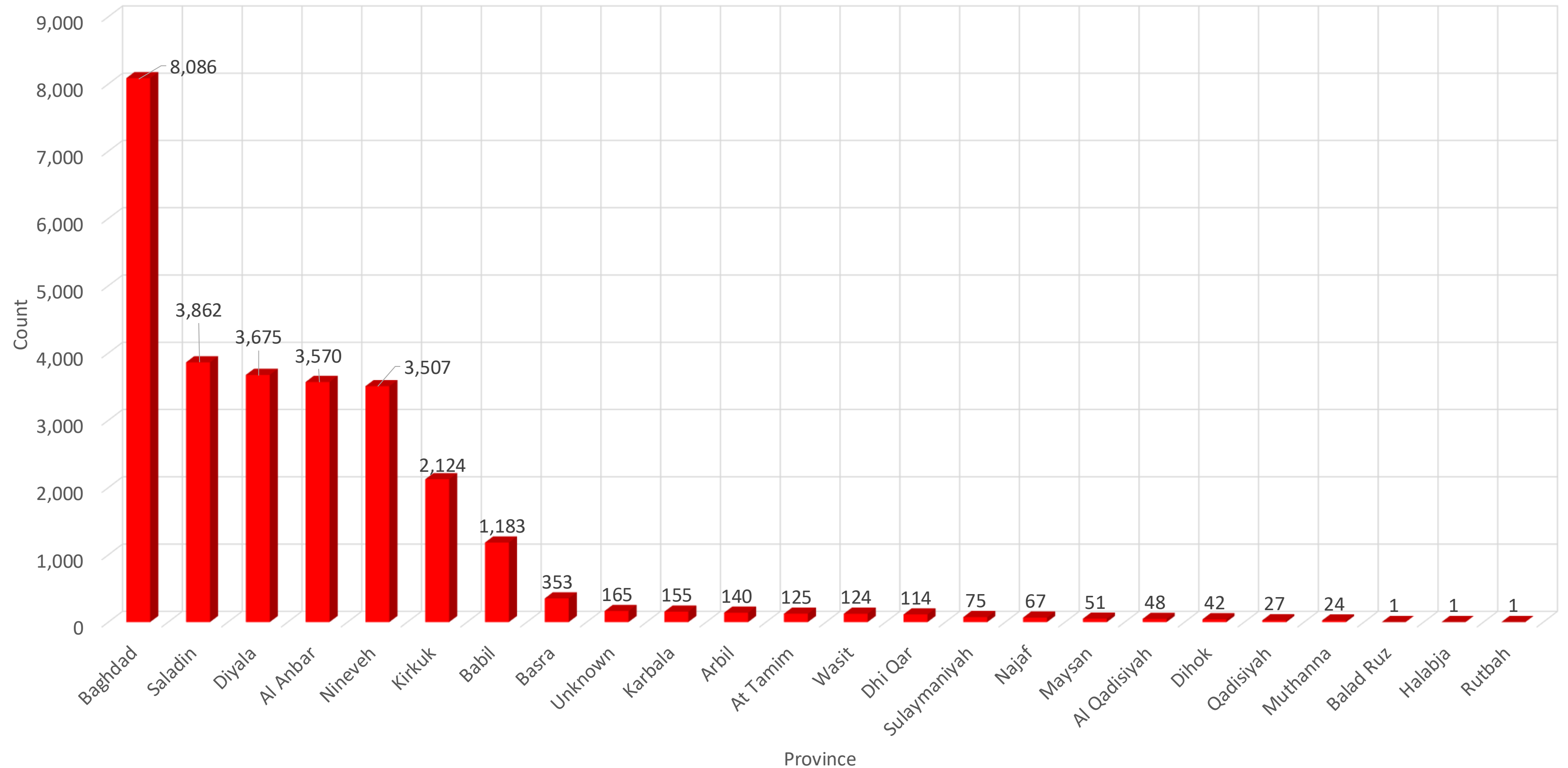
What are the main reasons of terrorism in Iraq?

- Political instability and weak government
 - The government struggles to keep control after the war which created chaos and power gaps that terrorist groups can exploit
- Sectarian divisions (Sunni vs. Shia)
 - Long lasting religious tensions led to violence and division which extremist groups use to justify attacks and recruit members
- Extremists' groups
 - These groups took advantage of the instability to gain territory and spread propaganda and to carry out attacks.
- Foreign involvement and conflict spillover
 - Military and political actions by other countries sometimes increased tensions and fueled resentment among local populations
- Economic problems
 - Limited job opportunities make some individuals more valuable to recruitment by terrorist groups offering money and purpose
- Weak security and military challenges
 - In certain periods and regions Iraq military and police were not strong enough to prevent terrorist activity
- Social grievances and lack of trust in government
 - Some groups feel excluded from economic power and terrorist groups exploit these grievances to gain support



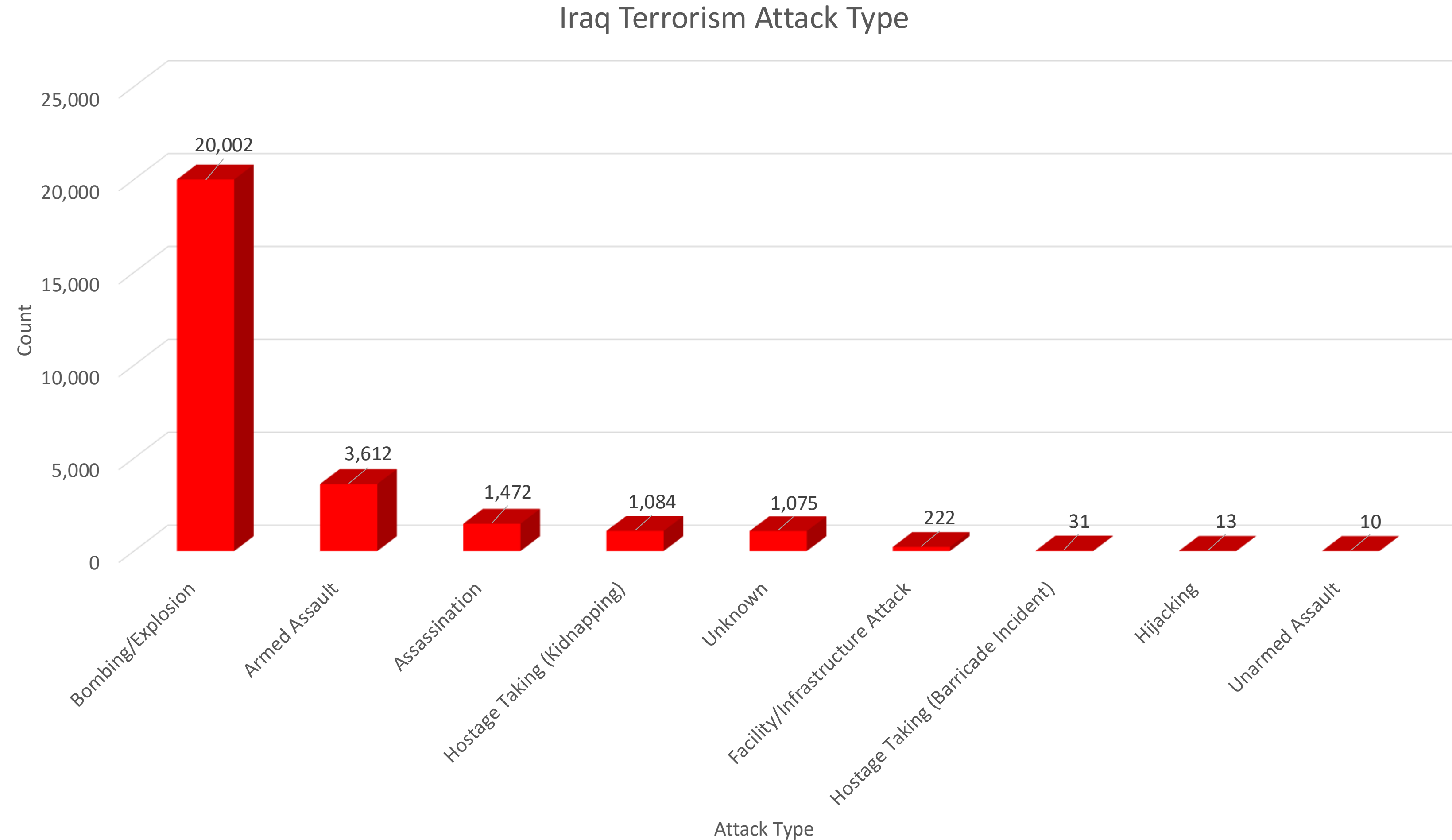
Show me the distribution across provinces

Iraq Terrorism Attacks Across All Provinces





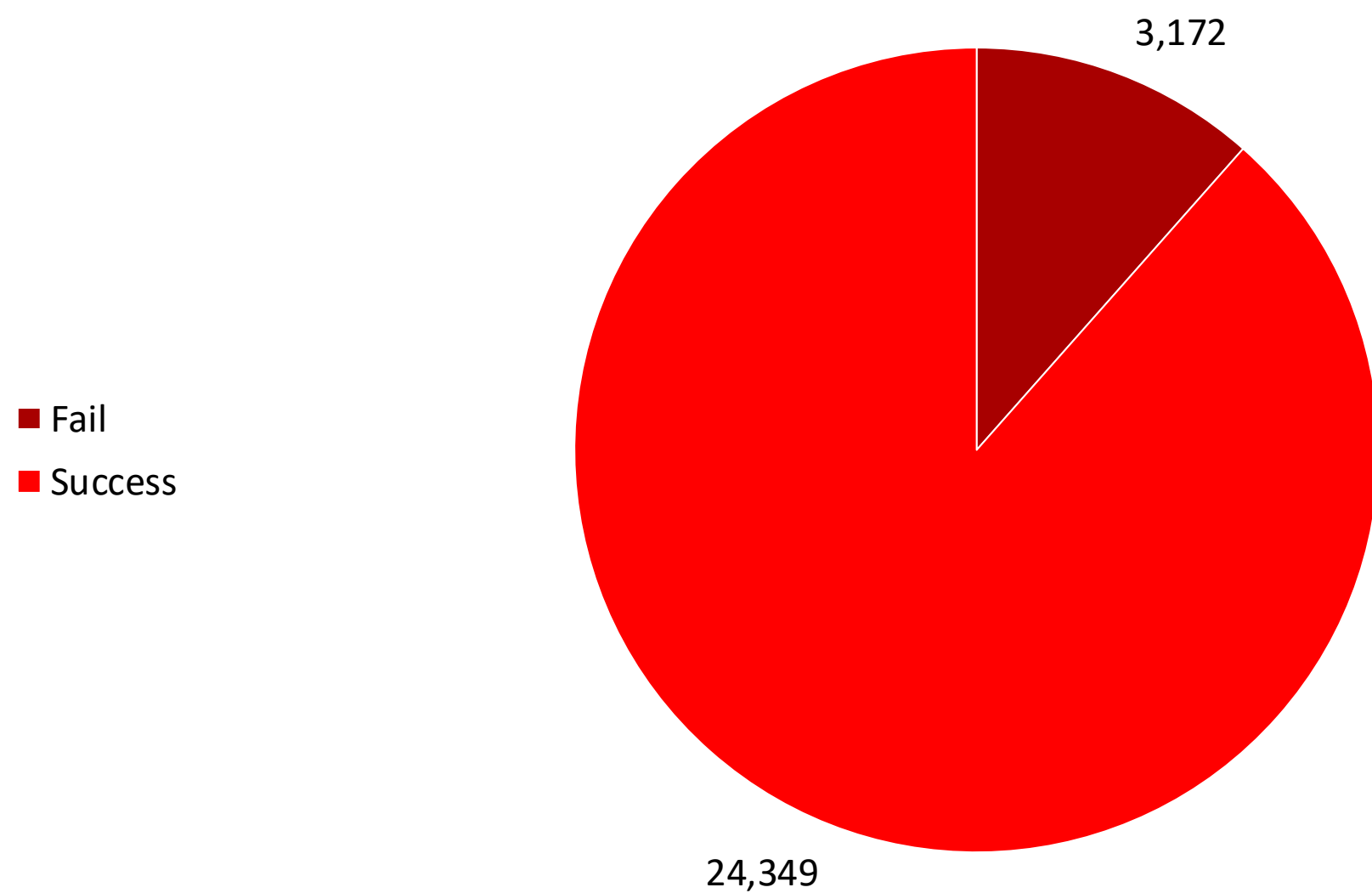
What types of attacks were conducted by the terrorists?



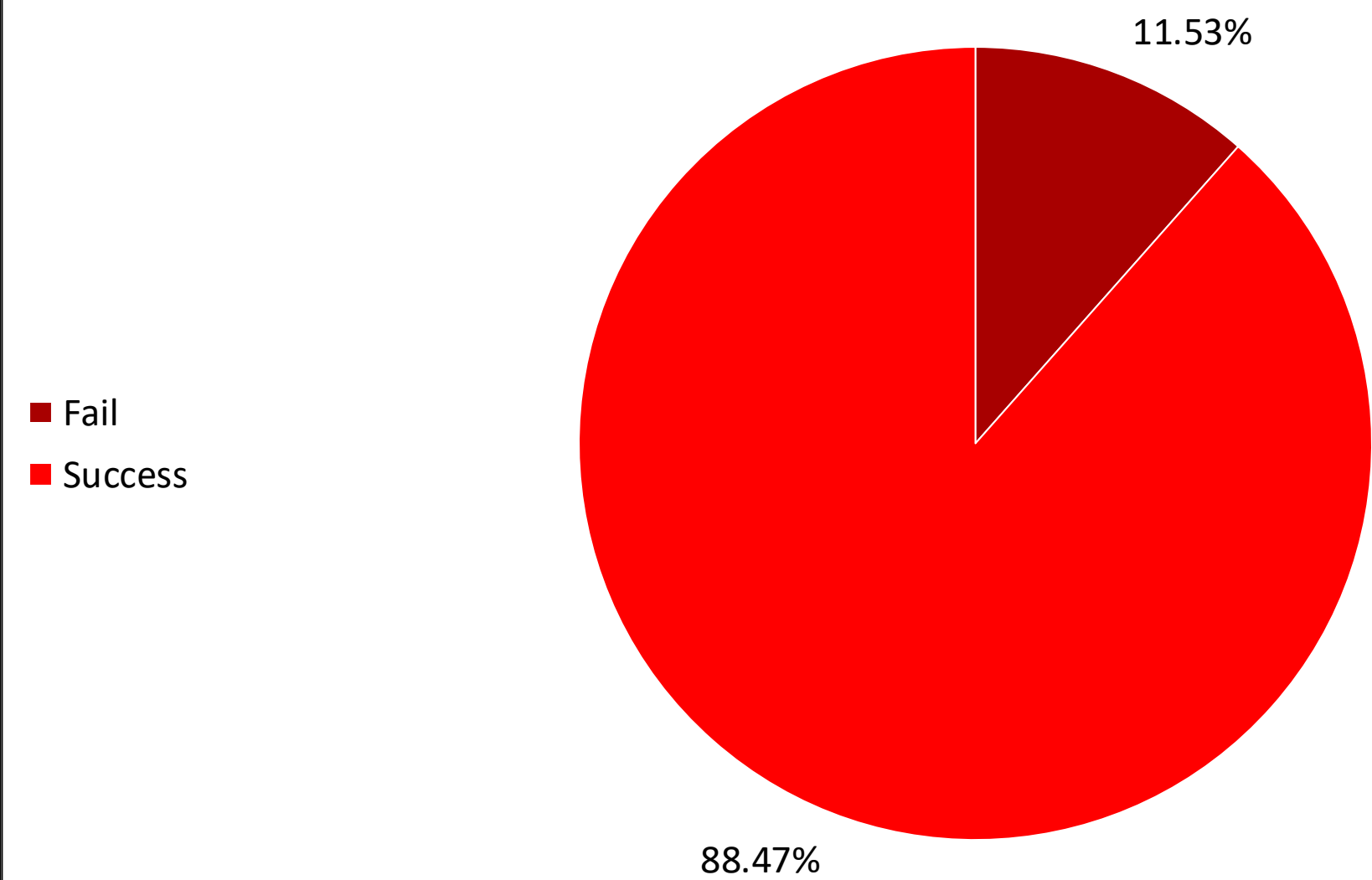


What was the success rate for terrorist attacks

Iraq Terrorism Success Rate Count



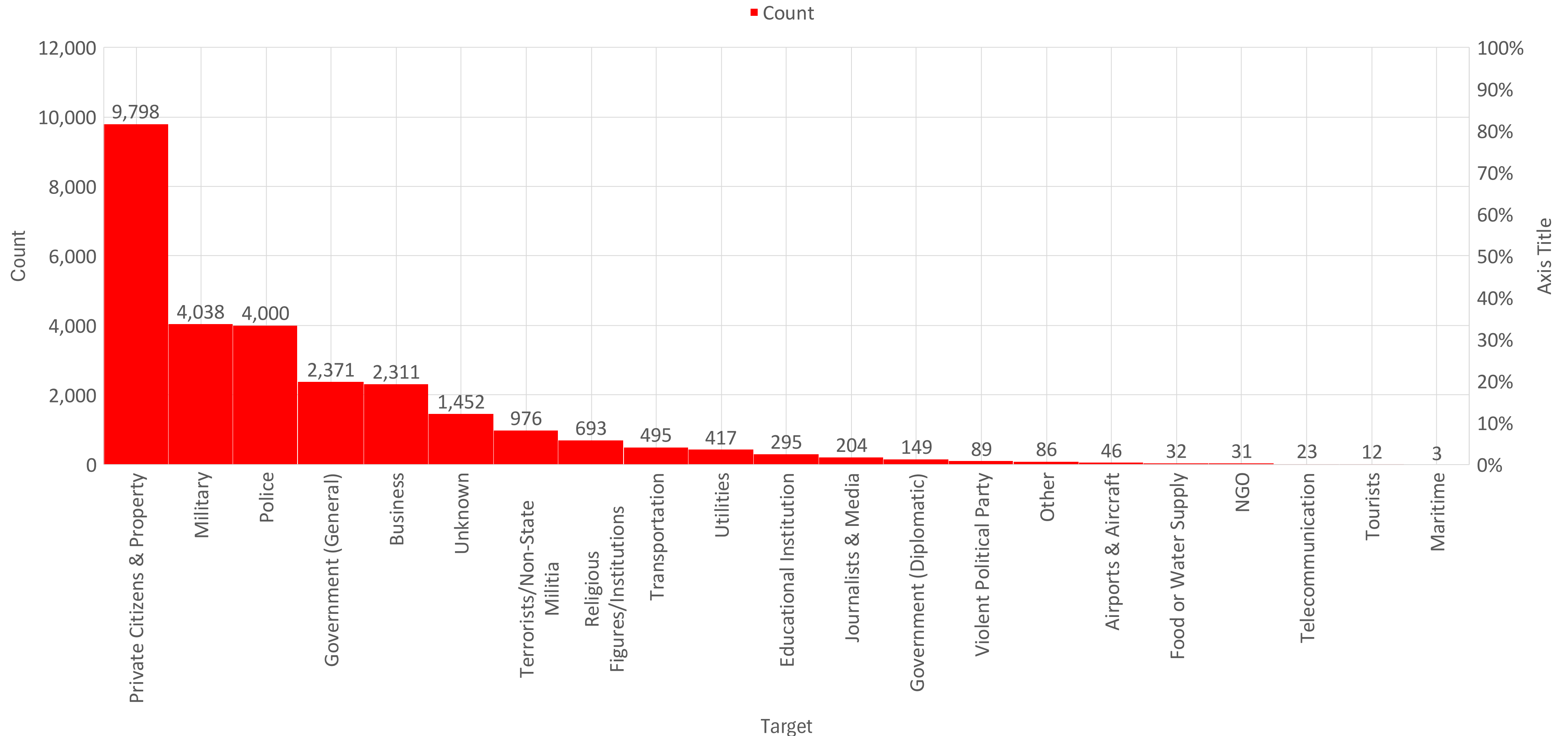
Iraq Terrorism Success Rate Percentage





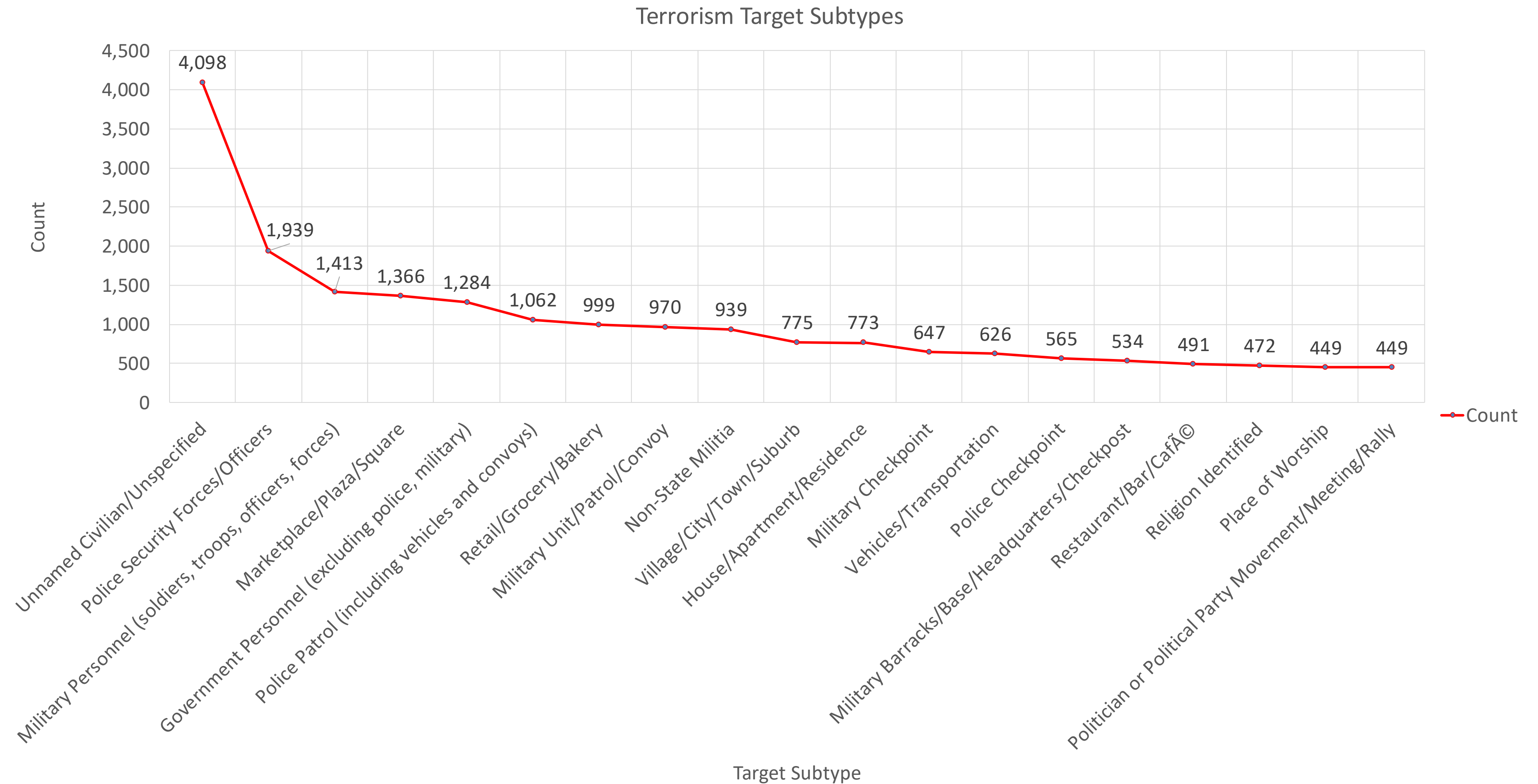
Who were targeted by these attacks?

Iraq Terrorism Events By Target Type





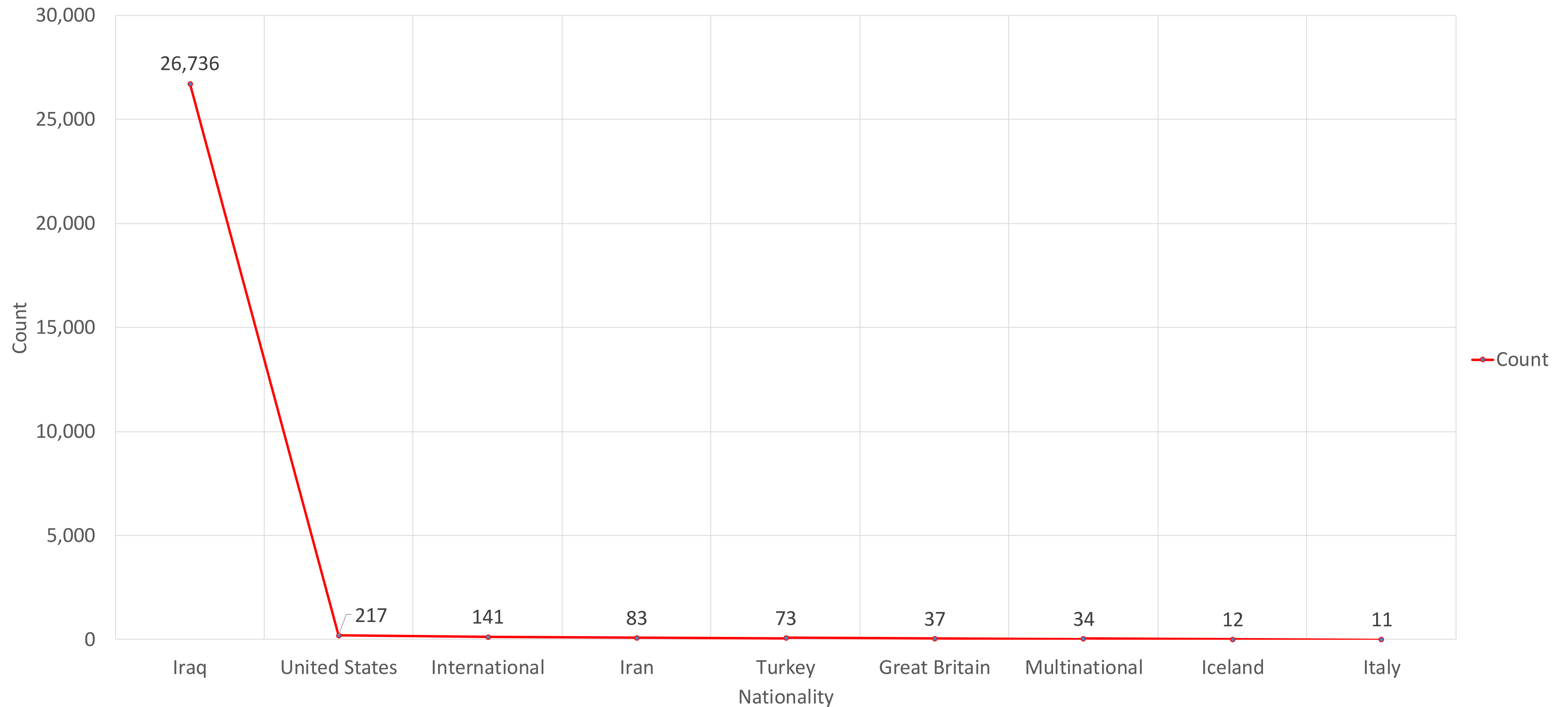
What were the target subtypes?





What were the nationalities of the targets?

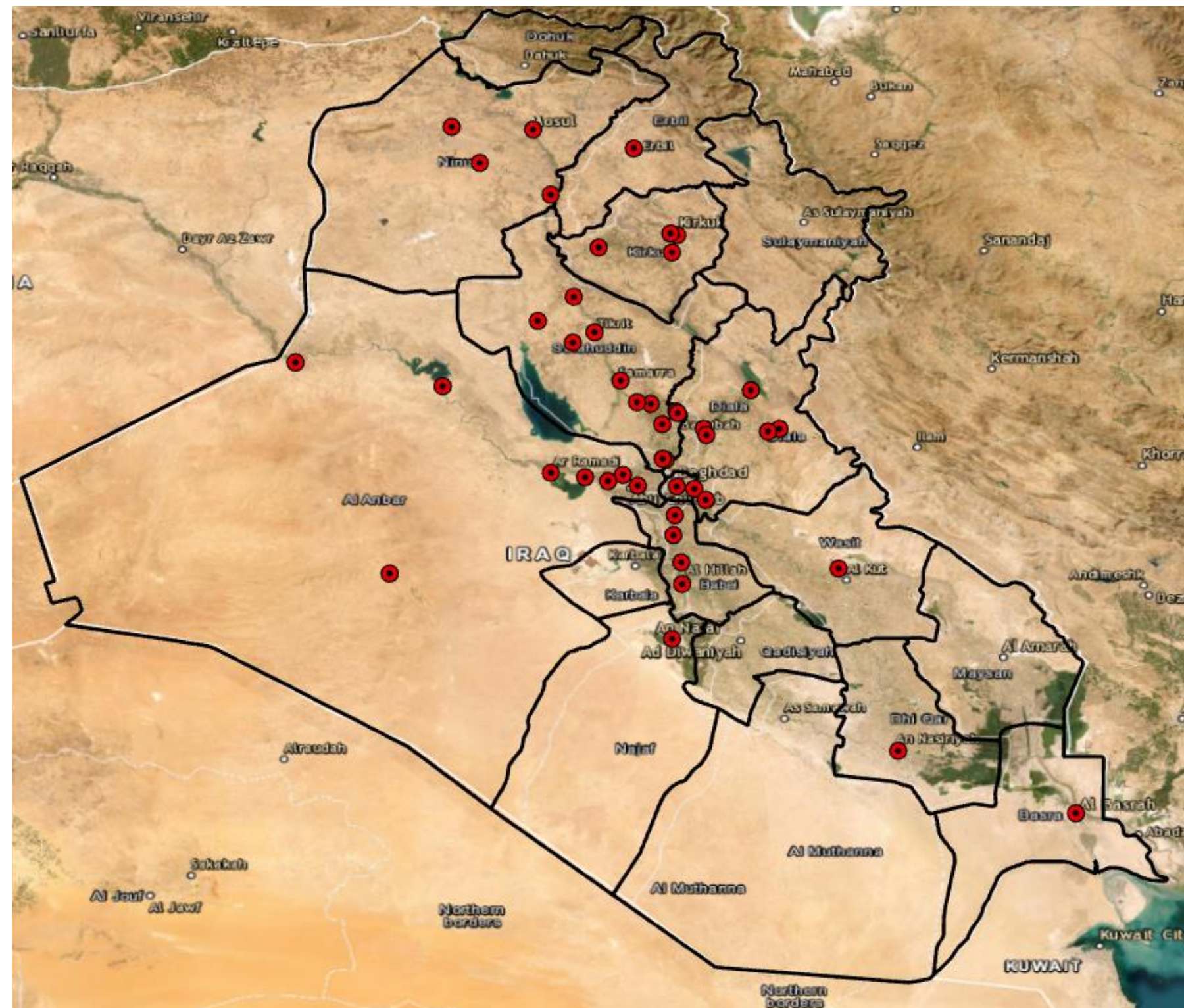
Iraq Terrorism Events By Nationality





Show me the incidents that targeted in US nationals?

217



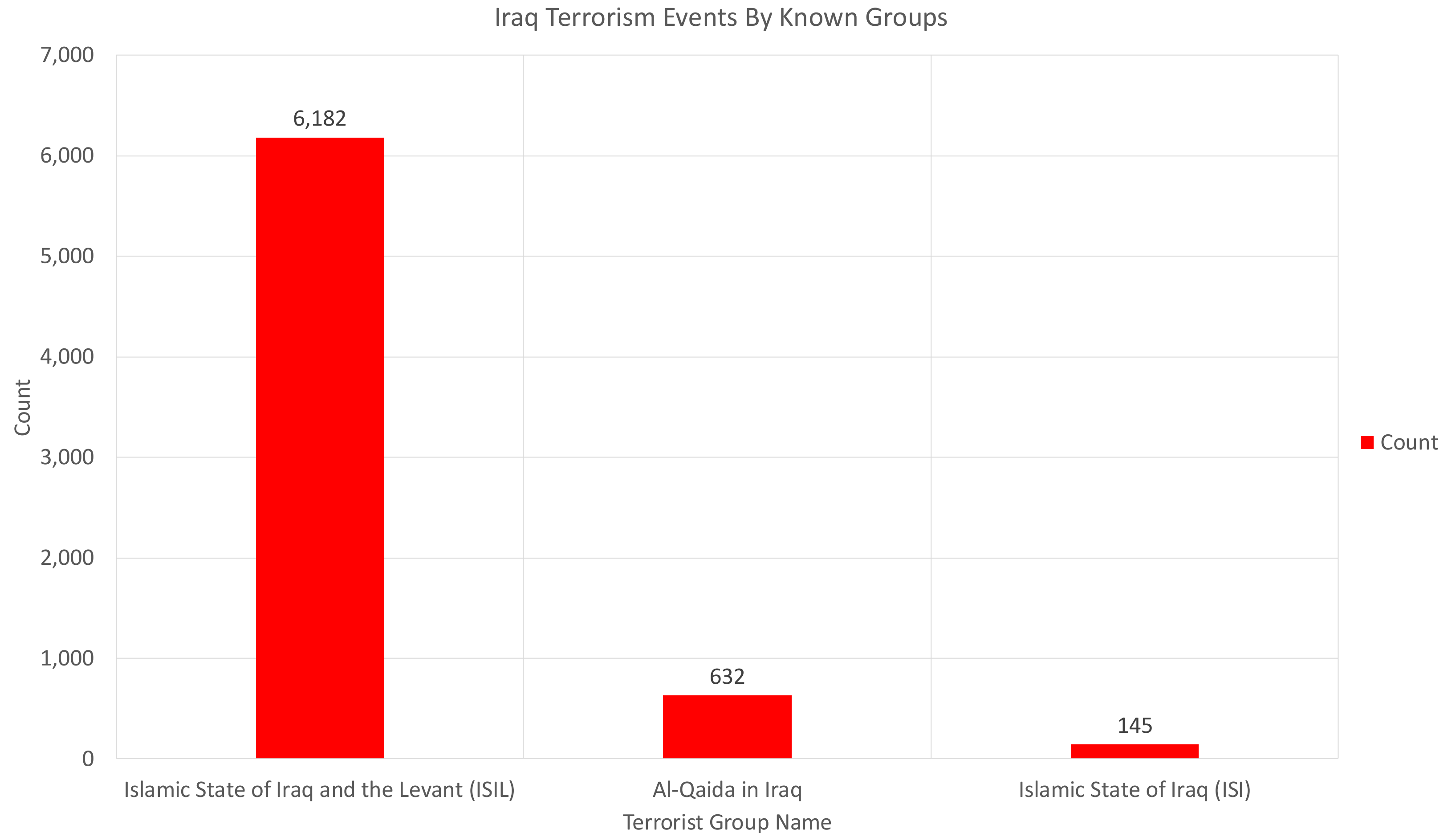


Show me the attacks on US Diplomats?

4



What terrorist groups carried out the attacks?





Islamic State of Iraq and the Levant(ISIL)

- ISIL (ISIS/Daesh) is a militant extremist group that emerged from al-Qaeda in Iraq and operates mainly in Iraq and Syria.
- Their ideology is a brutal, fundamentalist Sunni jihadist ideology that uses extreme violence, public executions, and destruction of heritage sites to enforce its rule.
- They lost most of their territory in Iraq and Syria by 2019, but they continue to operate as a terror threat via underground cells and global affiliates.



Al-Qaeda

- Al-Qaeda is a militant extremist organization founded in 1988 by Osama bin Laden and others, originally emerging from fighters involved in the Soviet-Afghan War and later operating globally.
- Their ideology is a radical Sunni jihadist belief system that seeks to establish Islamic governance and opposes Western influence, often using coordinated terrorist attacks against civilian and military targets.
- The group carried out major attacks such as the September 11 attacks, and although weakened by counterterrorism efforts, it continues to exist through regional affiliates and remains an ongoing global security threat.

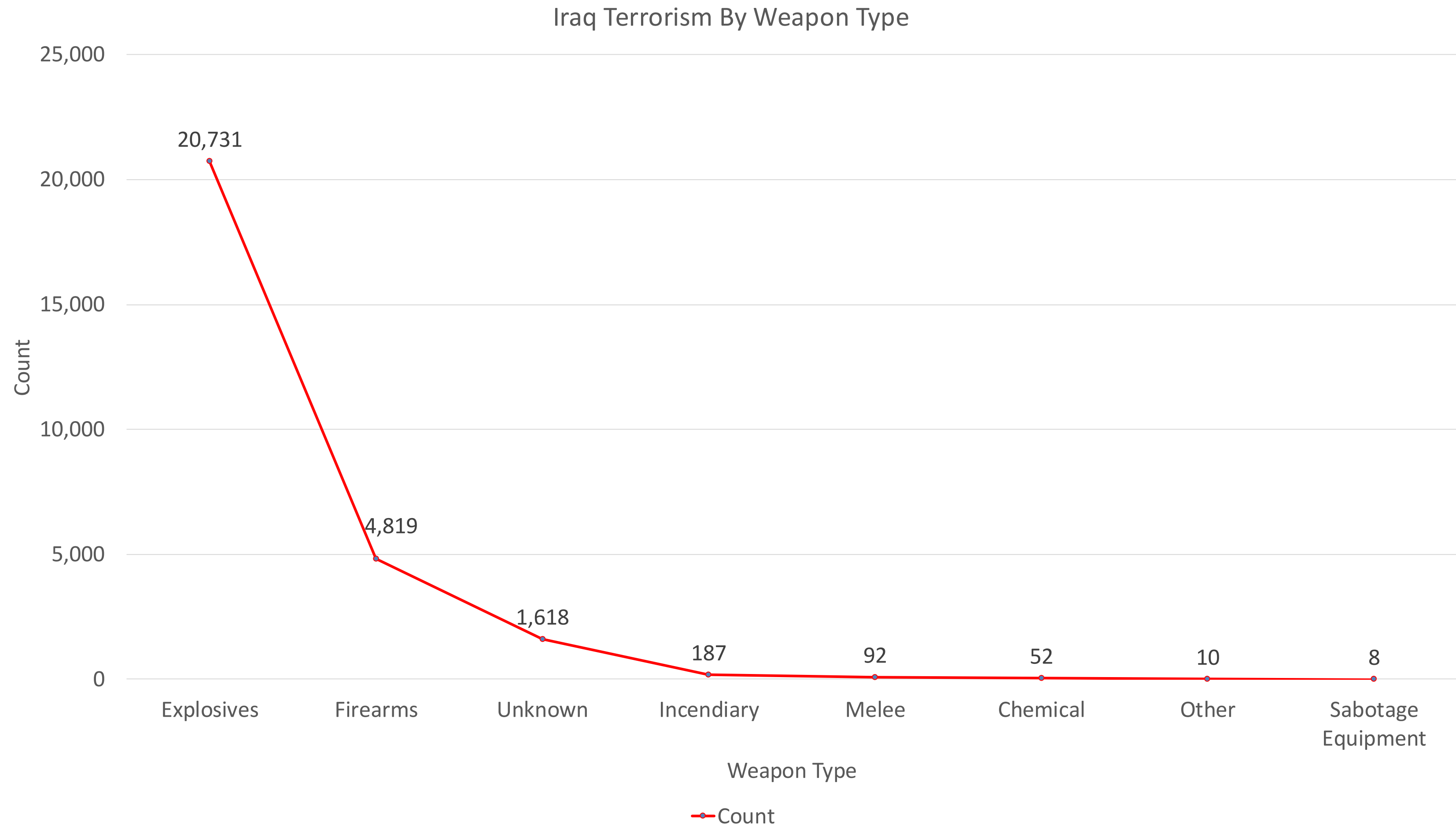


Islamic State of Iraq (ISI)

- The Islamic State of Iraq (ISI) was a militant extremist organization that emerged in 2006 as the successor to Al-Qaeda in Iraq after the death of its leader Abu Musab al-Zarqawi.
- Its leadership included figures such as Abu Omar al-Baghdadi and Abu Ayyub al-Masri, and it continued a Sunni jihadist ideology focused on establishing an Islamic state in Iraq through violent insurgency and terrorism.
- ISI operated primarily as an underground insurgent network during the Iraq War, conducting bombings, and assassinations, and it later evolved into what became the core of the modern Islamic State.



What types of weapons did they use?





How many people were killed at these attacks?

Range	Count	Midpoint
0–53.1	26,552	26.55
53.1–106.3	87	79.7
106.3–159.4	16	132.85
159.4–212.5	5	185.95
212.5–265.6	4	239.05
265.6–318.8	3	292.2
371.9–425	2	398.45
637.5–690.6	1	664.05
903.1–956.3	1	929.7
1646.9–1700	1	1673.45

Range: Kill range

Count: Instances where the kill count falls within that range

If we take the midpoint of each range, and multiply by the count (how many events with a kill count in that range) we can estimate the total number of kills to be 700k-720k+.

Example: $26,552 \times 26.55 = 704,955.6$



What were the deadliest attacks (list)?

1. 1,700 deaths

2. 953 deaths

3. 670 deaths



Deadliest attack (I)

- 1,700 deaths
 - Date: 2014
 - City: Tikrit, Iraq (Saladin Governorate)
 - Target: Iraqi Armed Forces Soldiers
 - Motive: Revenge for the killing ISIL leader Abdul-Rahman al-Beilawy.
 - Conducted by: Islamic State of Iraq and the Levant (ISIL)
 - Weapon: Firearms



Deadliest attack (II)

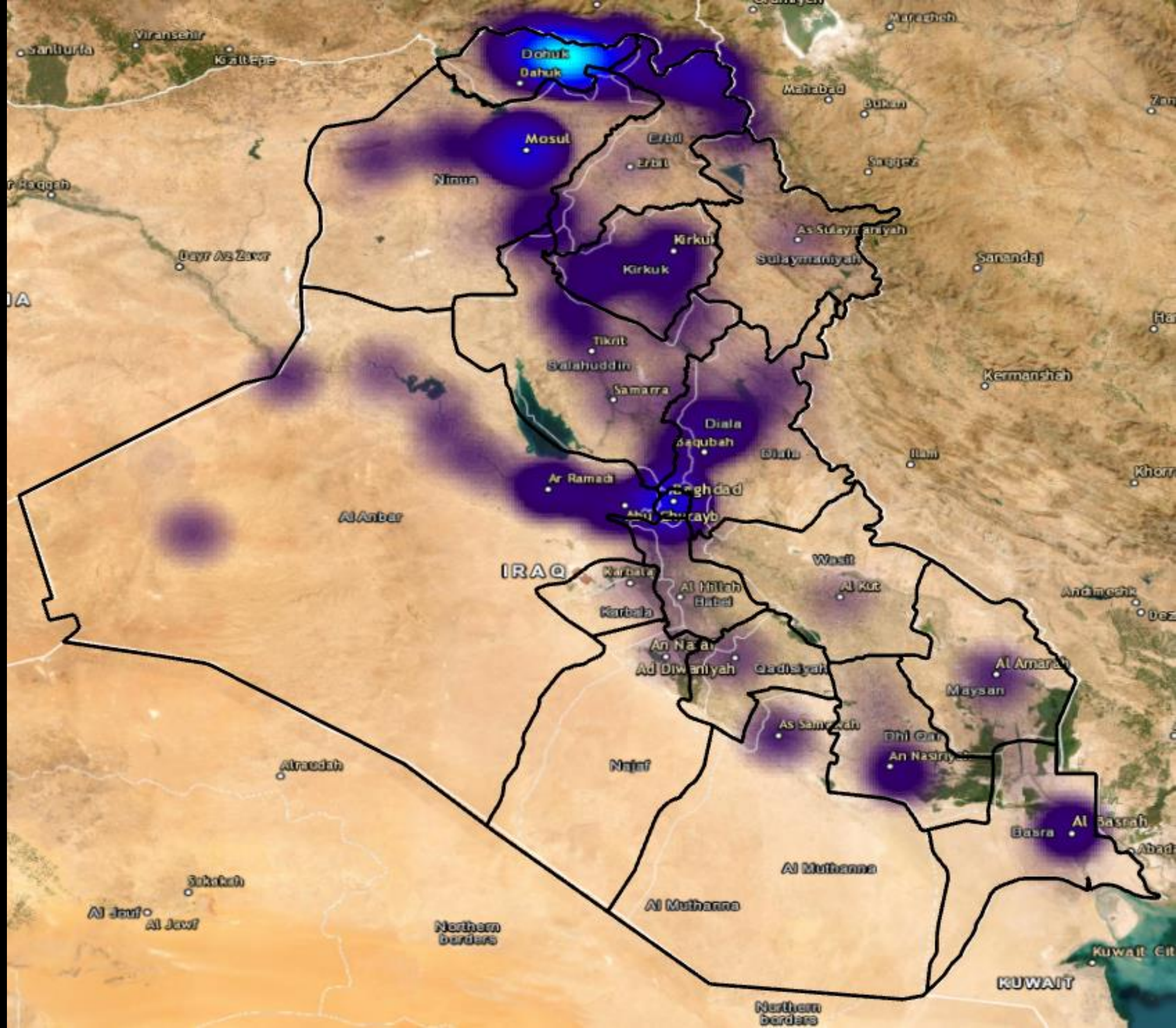
- 953 deaths
 - Date: 2014
 - City: Sinjar, Iraq (Nineveh Governorate)
 - Target: Yazidi Civilians
 - Motive: Force them to convert to Islam
 - Conducted by: Islamic State of Iraq and the Levant (ISIL)
 - Weapon: Unknown



Deadliest attack (III)

- 670 deaths
 - Date: 2014
 - City: Badush, Iraq (Nineveh Governorate)
 - Target: Iraq Prison Service, Badush Prison
 - Motive: Take over Mosul and liberate the people
 - Conducted by: Islamic State of Iraq and the Levant (ISIL)
 - Weapon: Projectile Explosives (rockets, mortars, RPGs, etc.)

Conflicts





What are the main reasons of conflicts in your country?

- Post-2003
 - The 2003 Iraq war removed Saddam Hussein's government
 - Security, military, and political collapsed
 - Created a power vacuum that allowed armed groups to form
- Sectarian division (Sunni vs. Shia)
 - Long standing religious split between Sunni and Shia Muslims
 - After 2003, political power shifted toward Shia-majority rule
 - Sunni communities often felt excluded, increasing tensions and insurgency
- Economic struggles
 - High unemployment, especially among youth
 - Oil dependent economy with unequal wealth distribution
 - Poverty makes recruitment by armed groups easier



Explain the violent incidents

Top 3 Violent Incidents

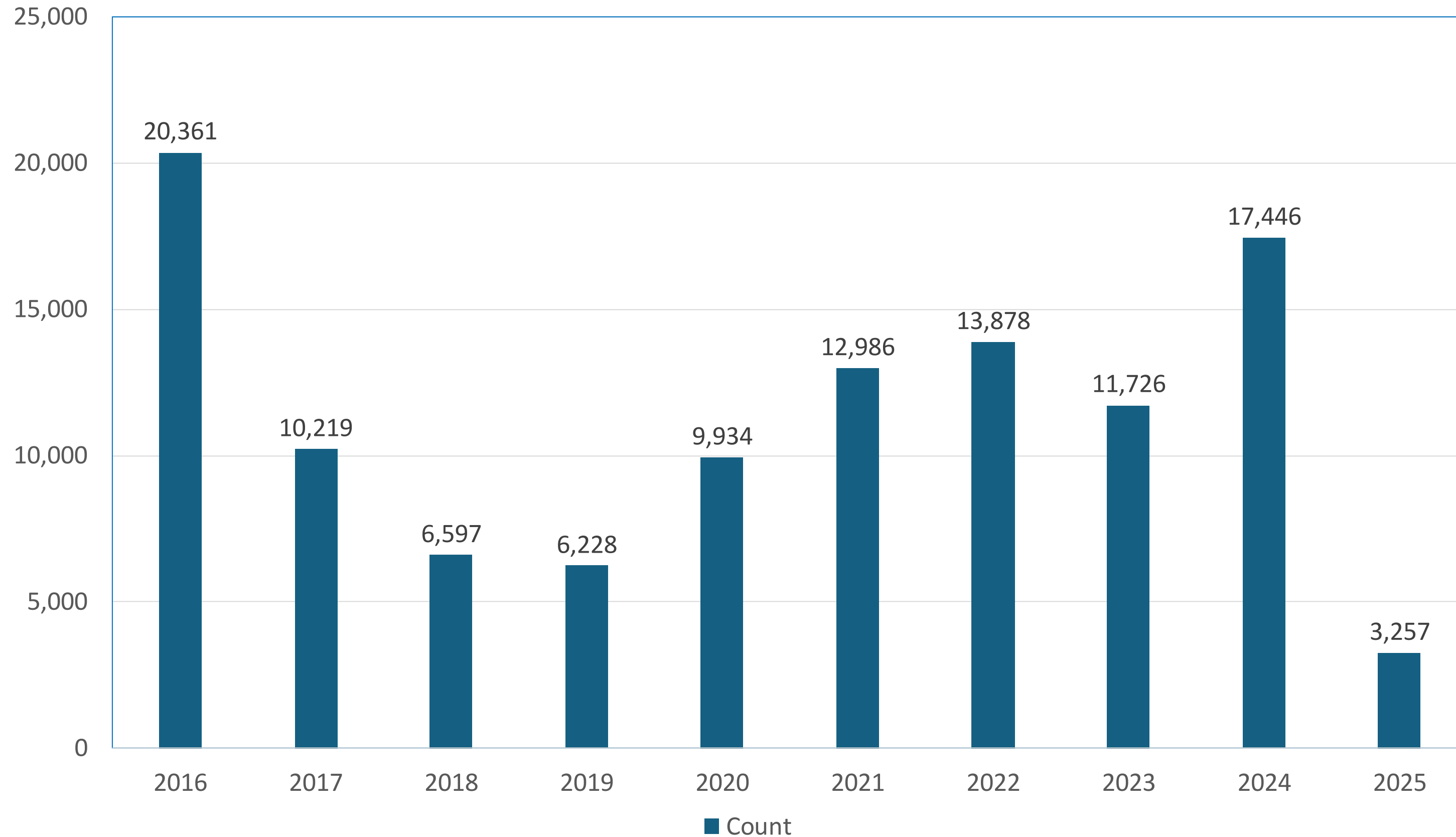
Types:

- | | | |
|--|--|---|
| <ul style="list-style-type: none">• State forces-rebel group• Iraqi Liberation• Fatalities: 600• July 8, 2016• Locations<ul style="list-style-type: none">• Mintaqat Zankurah• Ramadi | <ul style="list-style-type: none">• Rebel group-Civilians• Firing Squad• Fatalities: 300• Feb 7, 2016• Location<ul style="list-style-type: none">• Mosul | <ul style="list-style-type: none">• Rebel group-Civilians• Suicide Bomber• Fatalities: 293• July 2, 2016• Location<ul style="list-style-type: none">• Baghdad - Karadah |
|--|--|---|



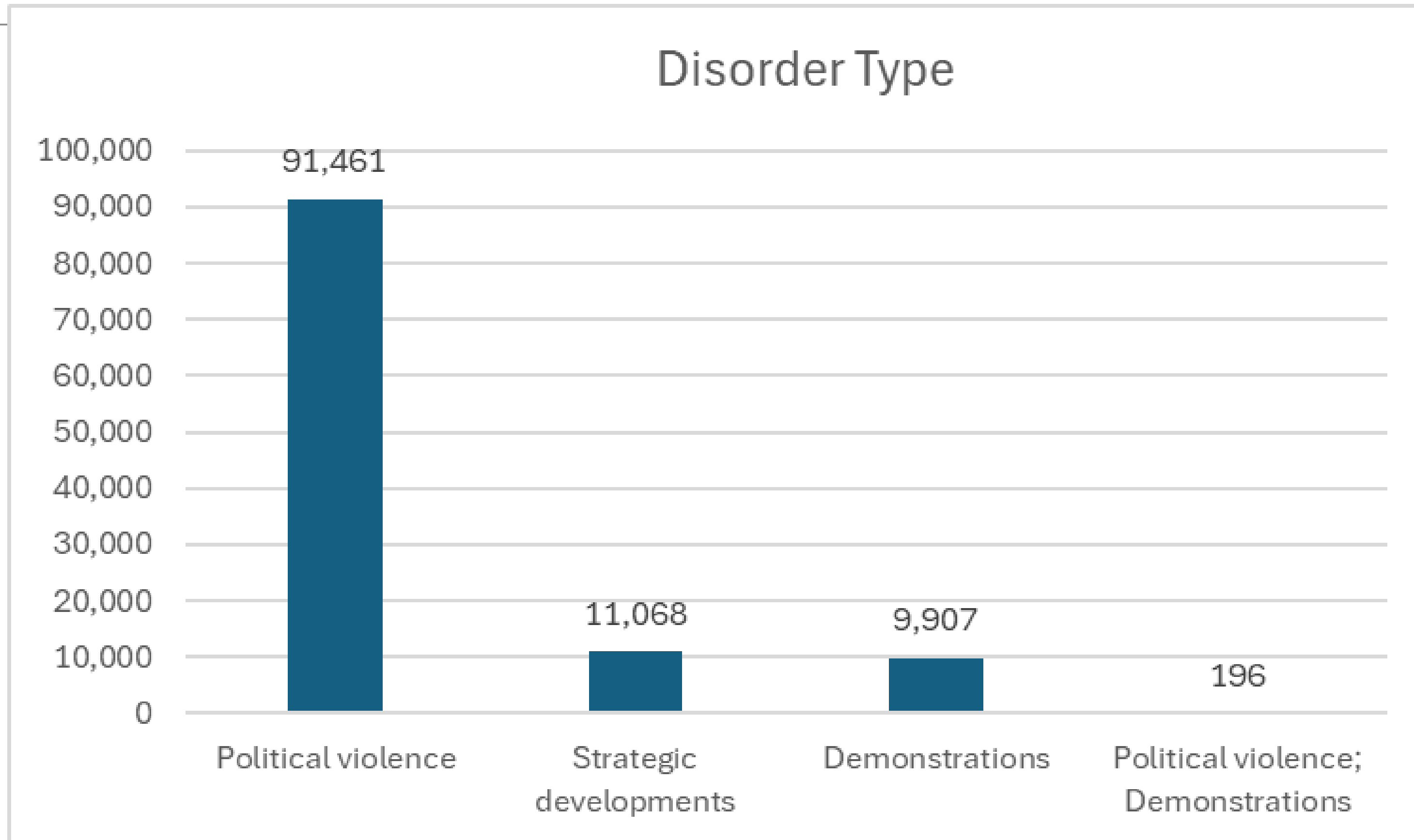
Show me the evolution of conflicts by year

Conflicts by Year



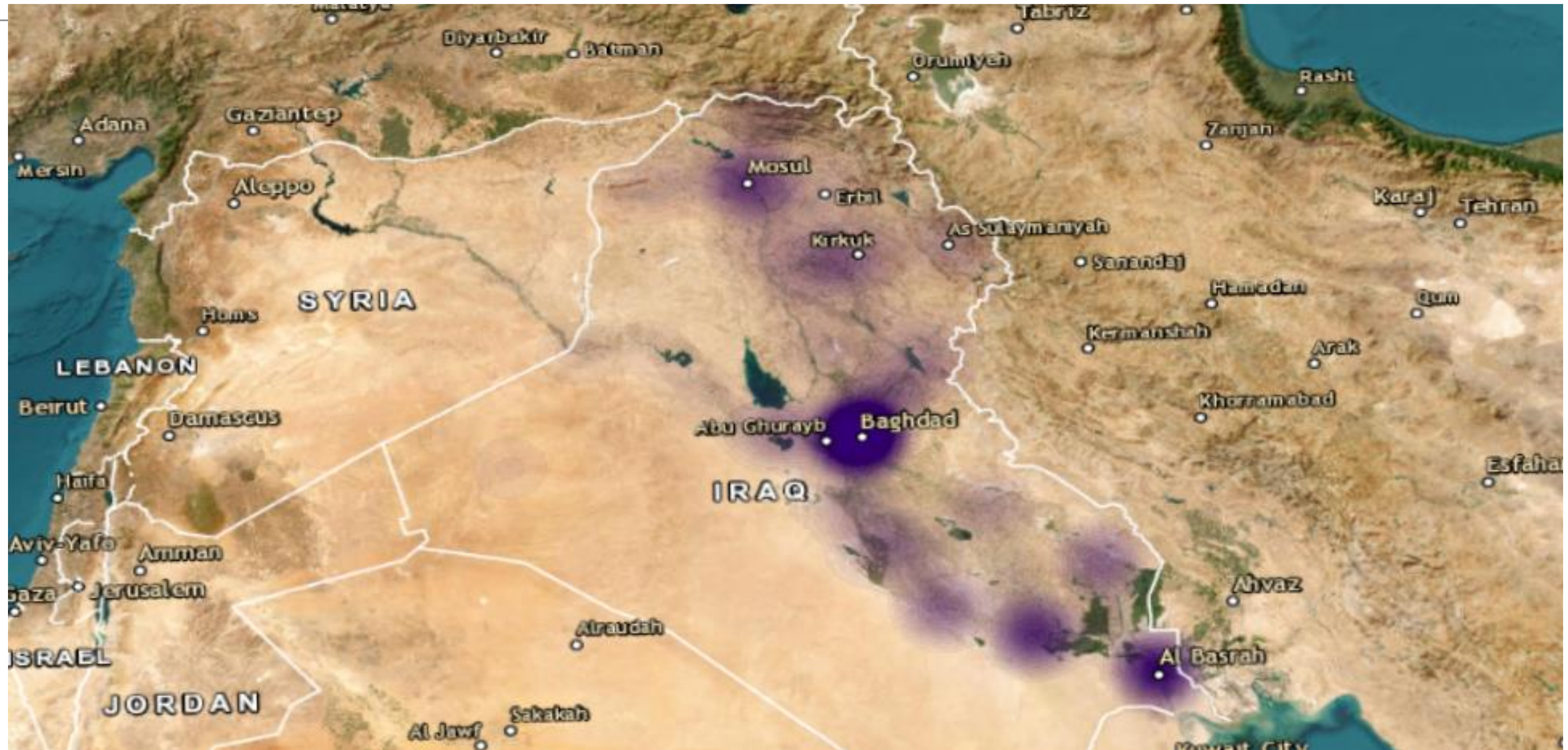


How was the disorder type distributed graphic?



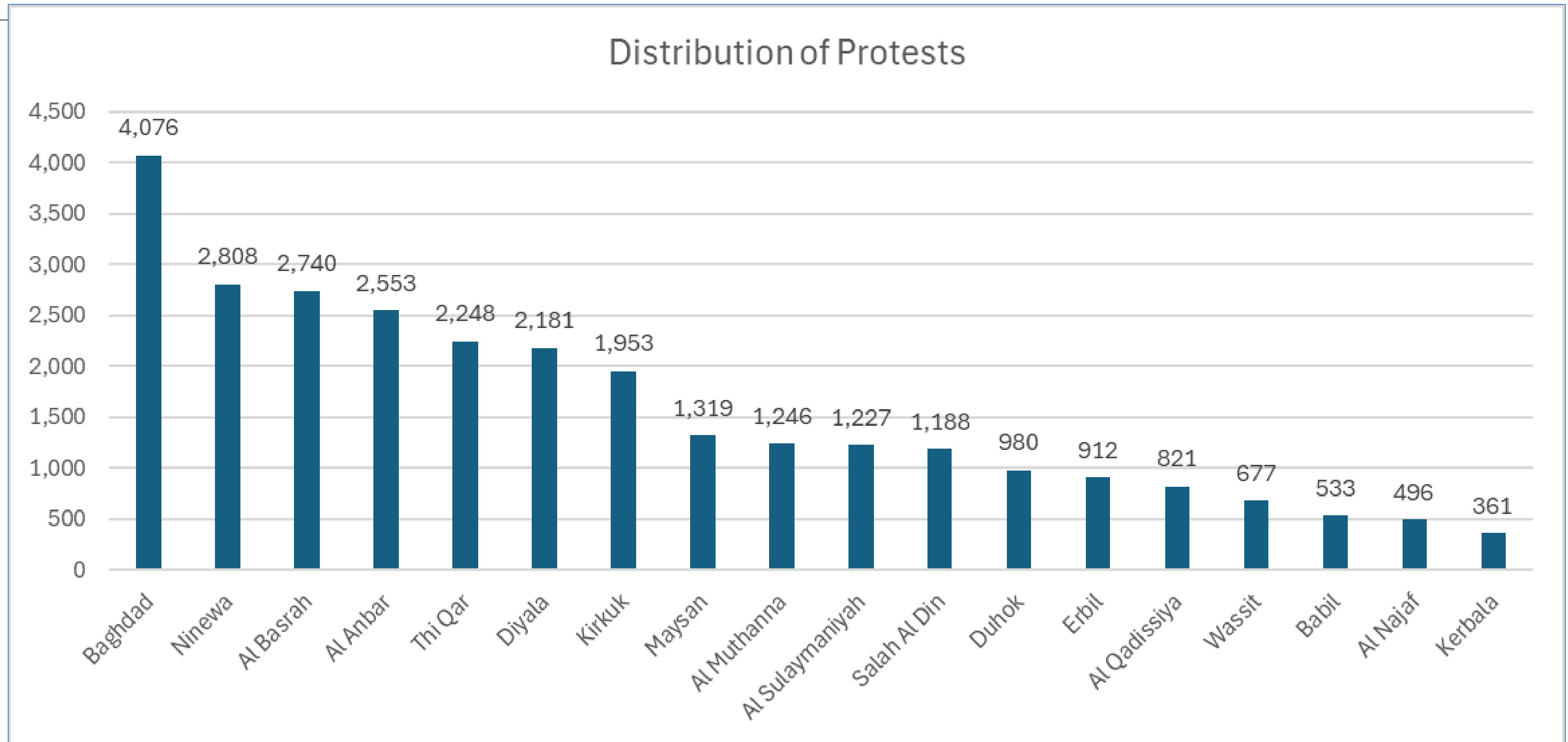


How were the protests distributed?



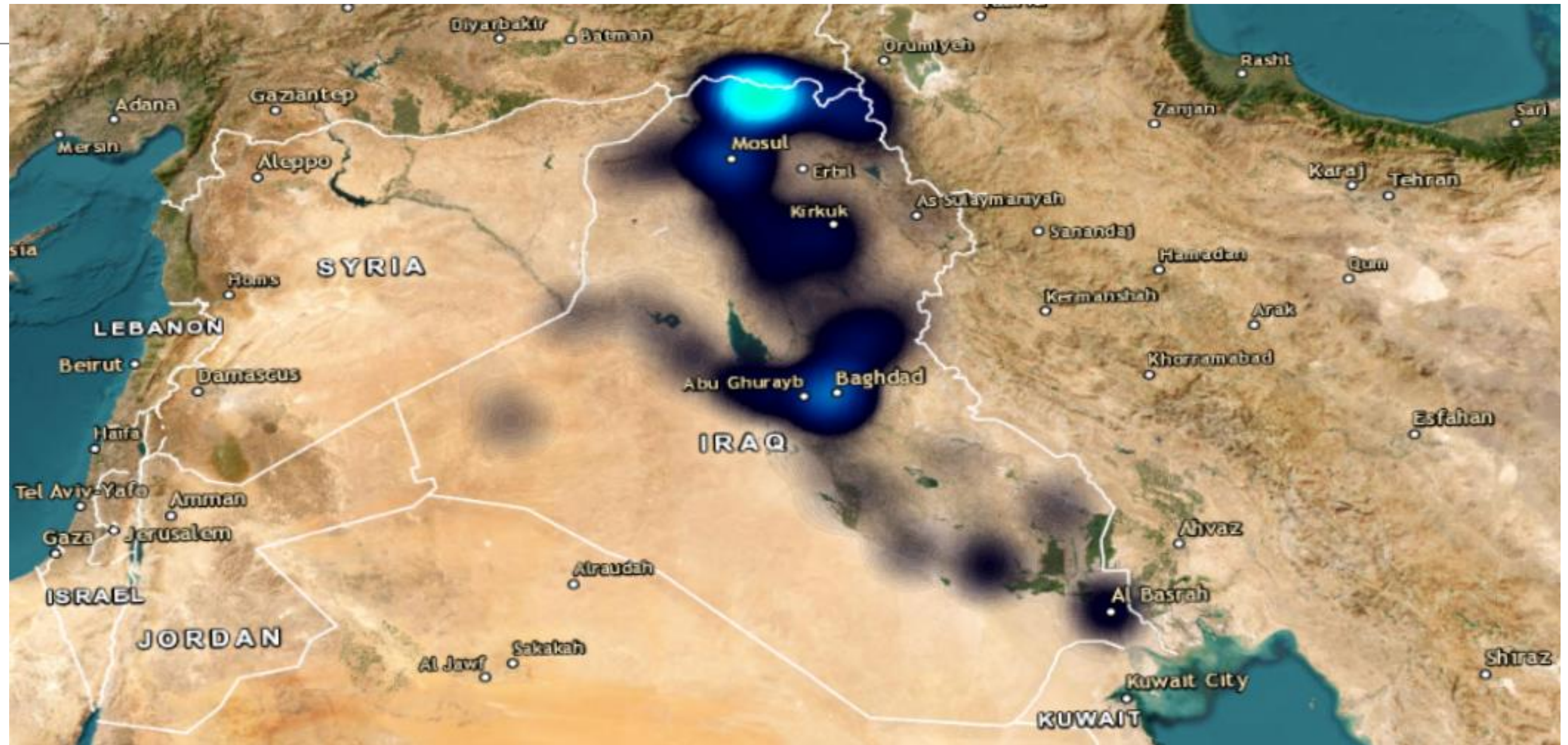


Distribution of protests graphic



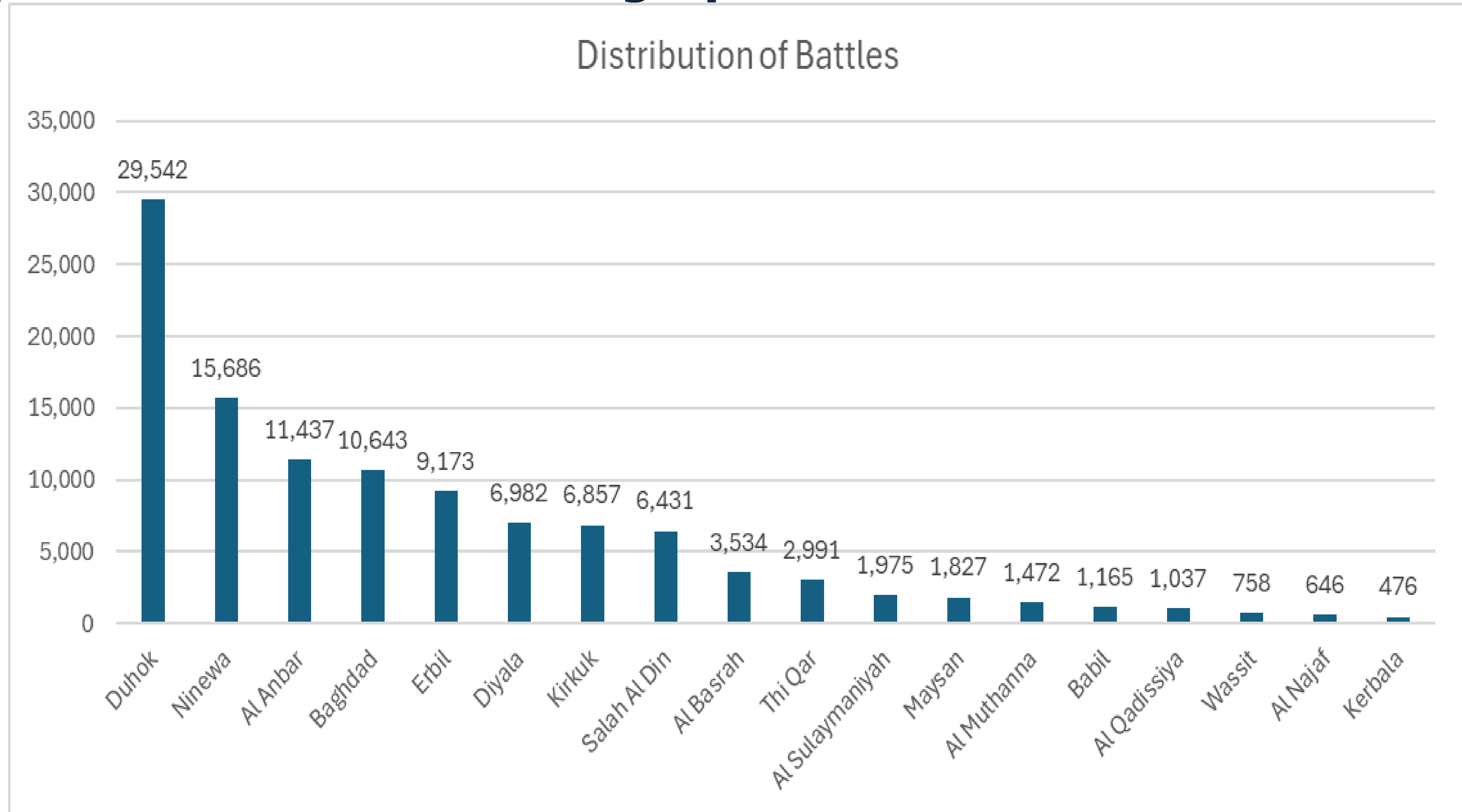


How were the battles distributed?



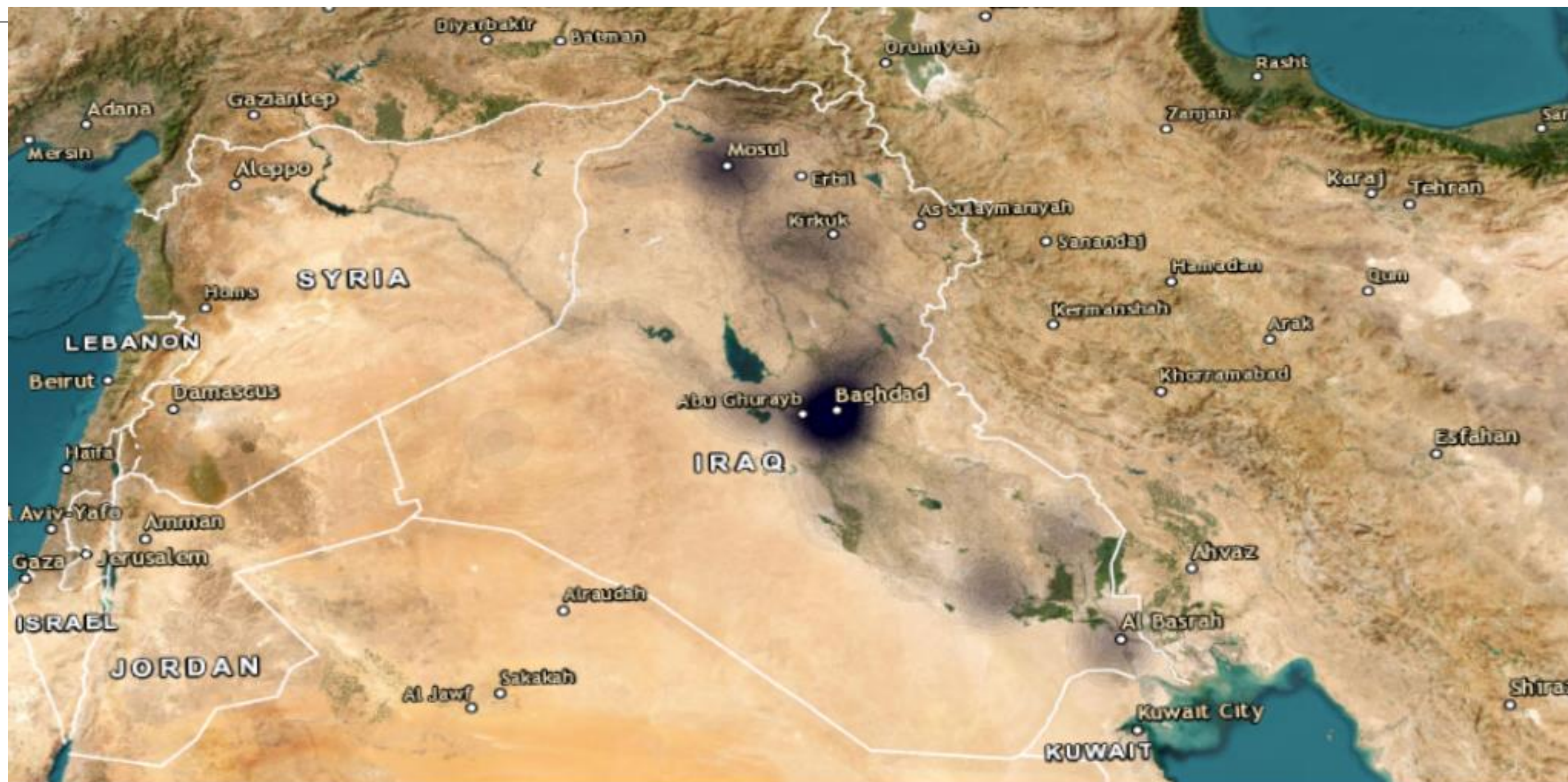


Distribution of battles graphic



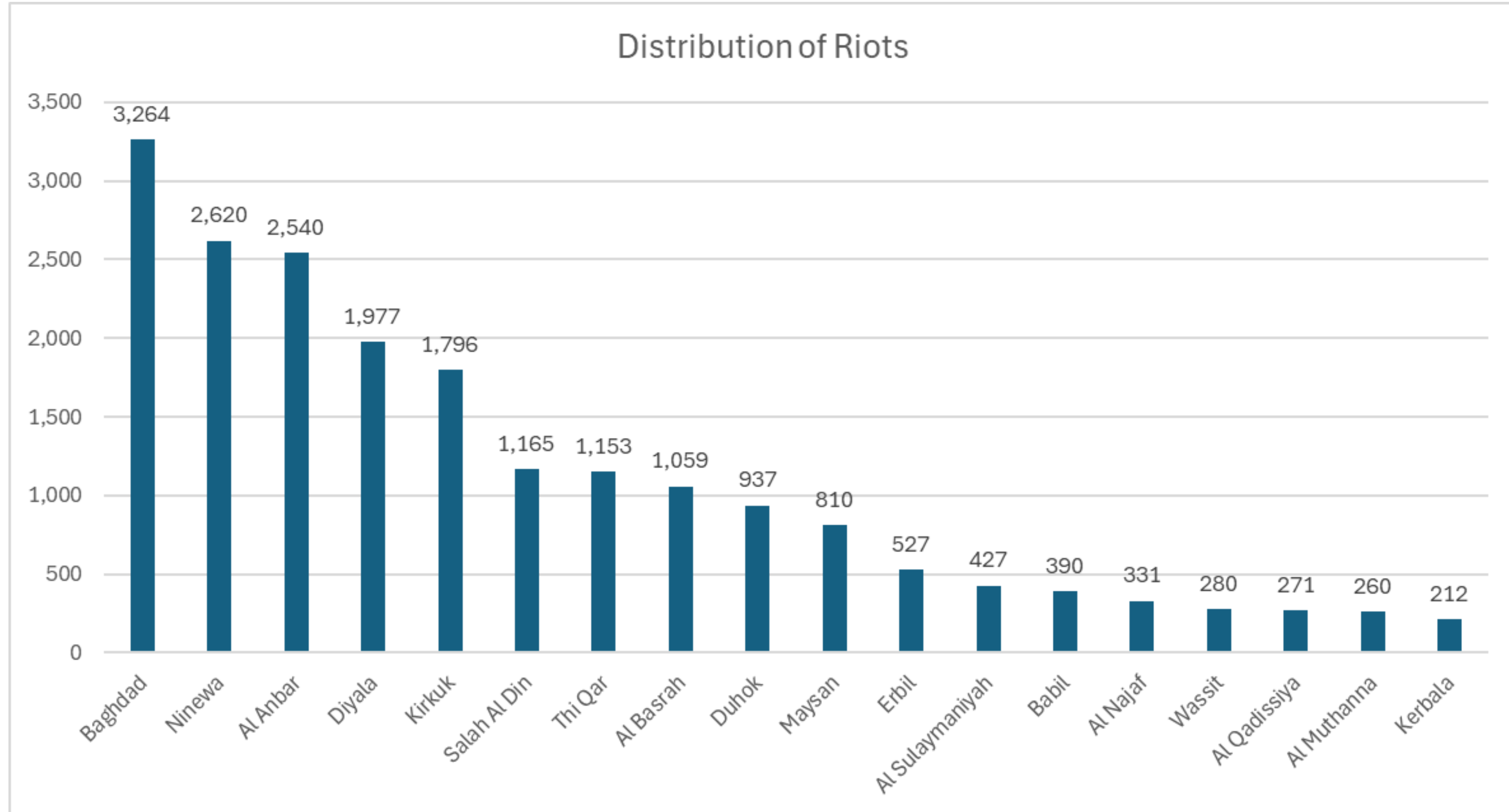


How were the riots distributed?



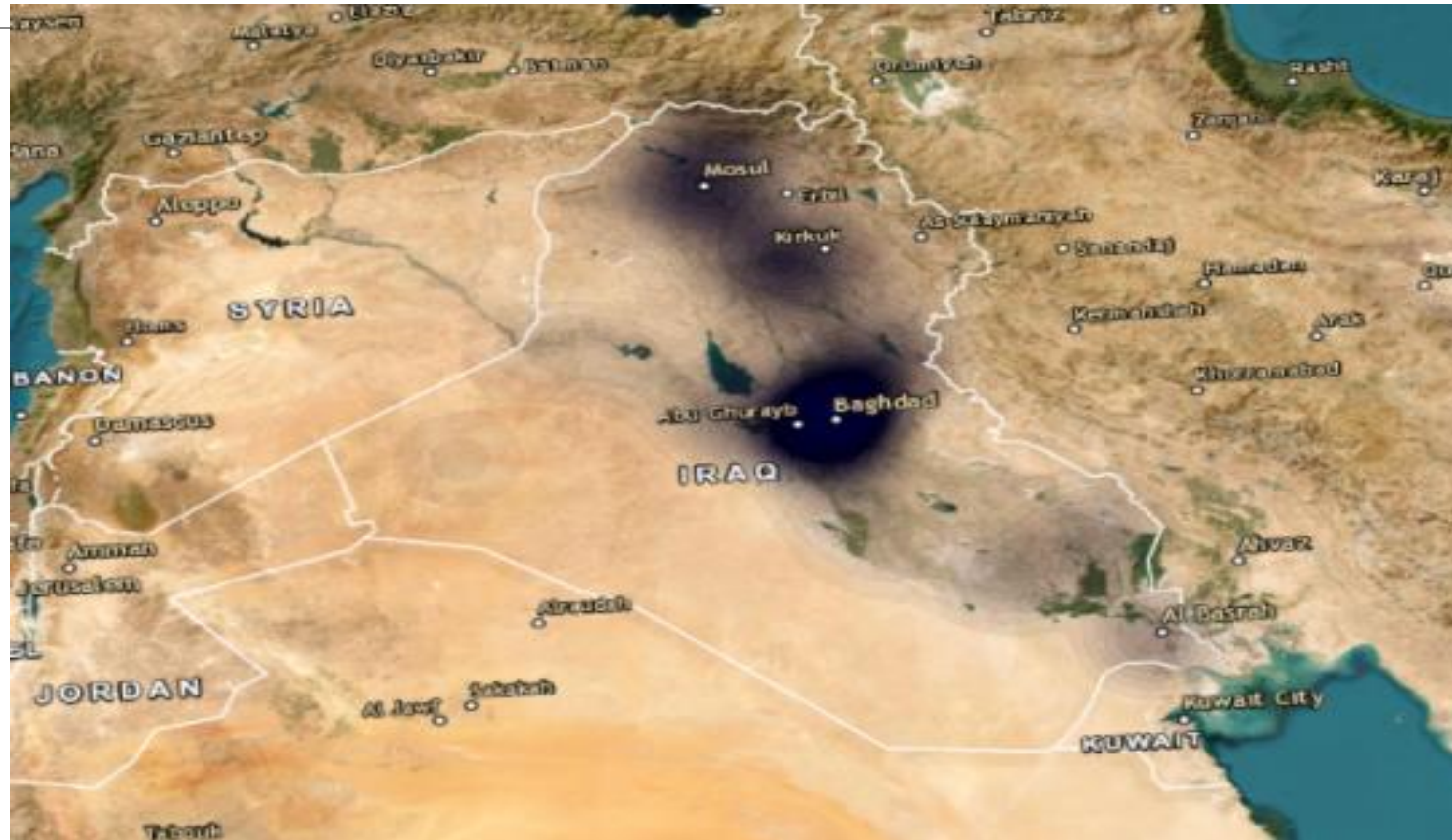


Distribution of riots graphic



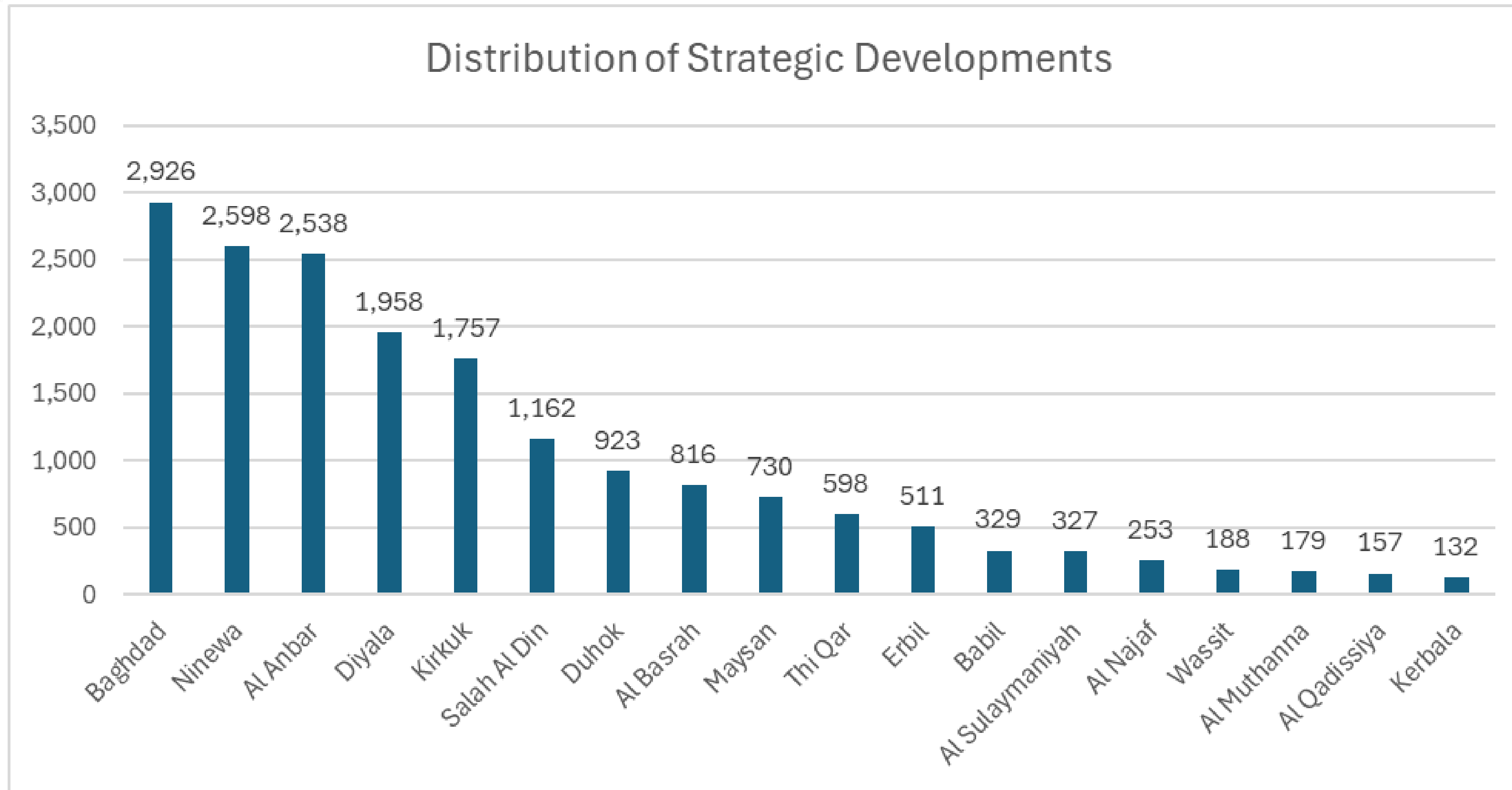


How were the strategic developments distributed?



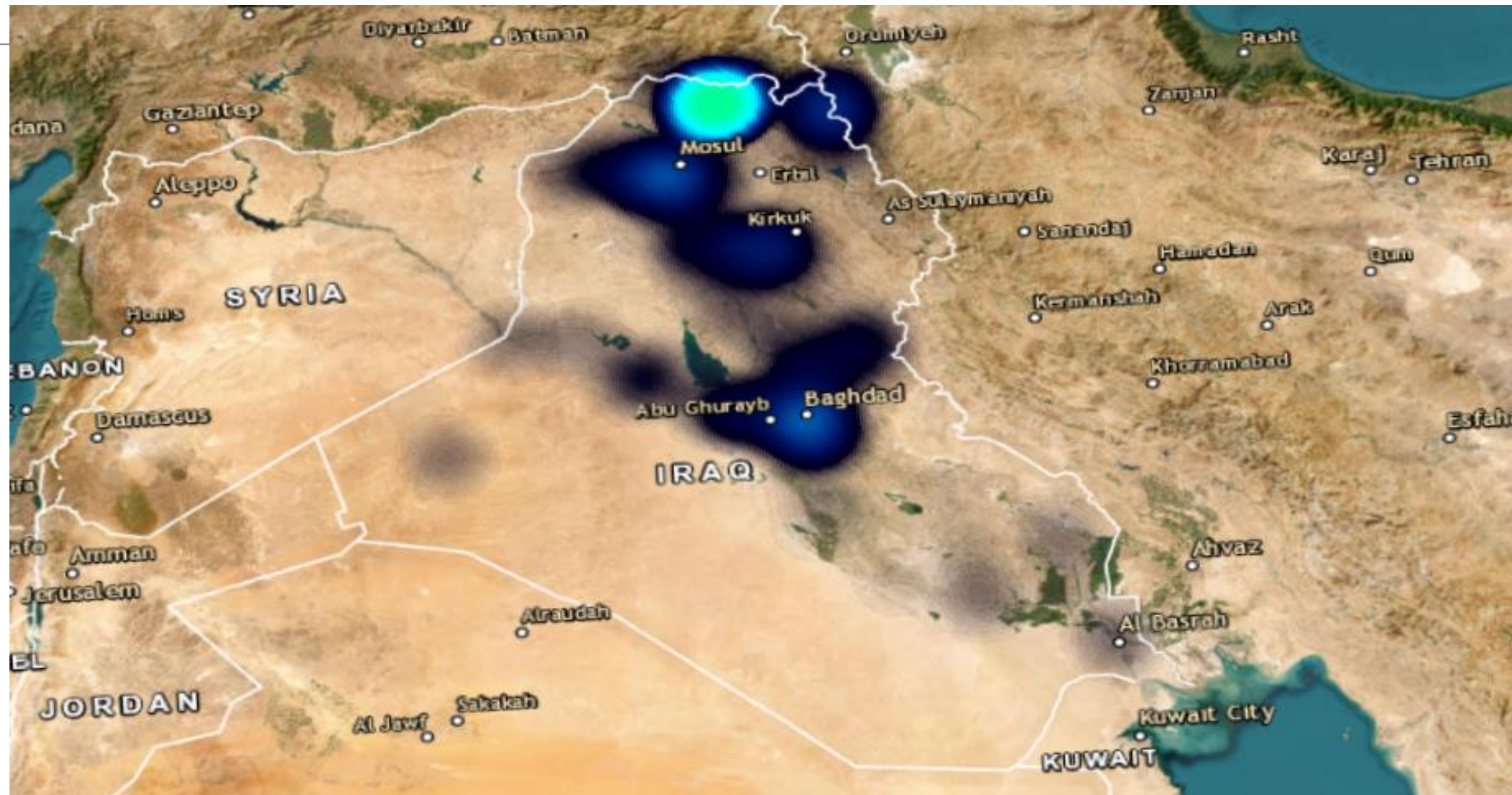


Distribution of strategic developments graphic



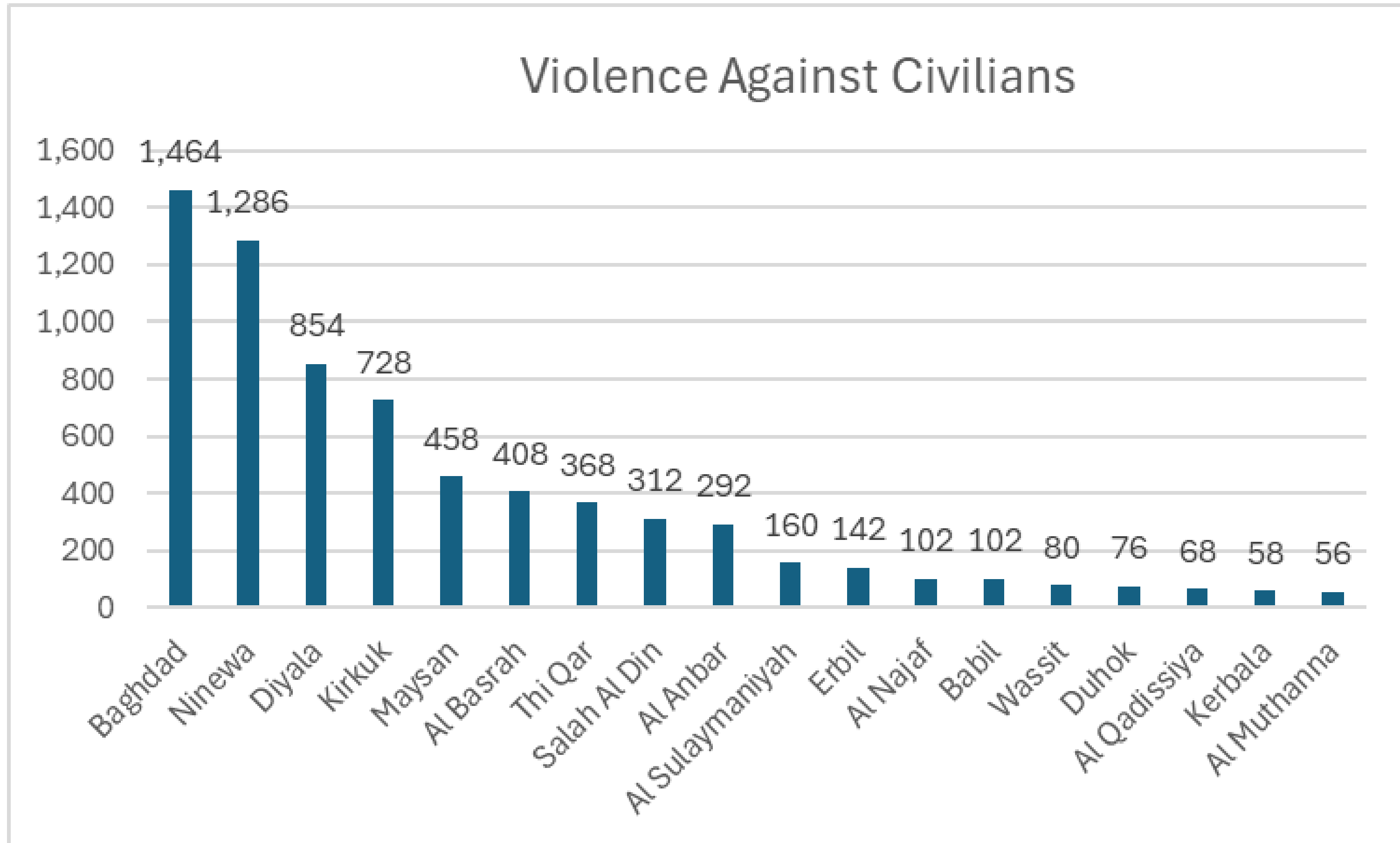


How were the violence against civilians distributed?



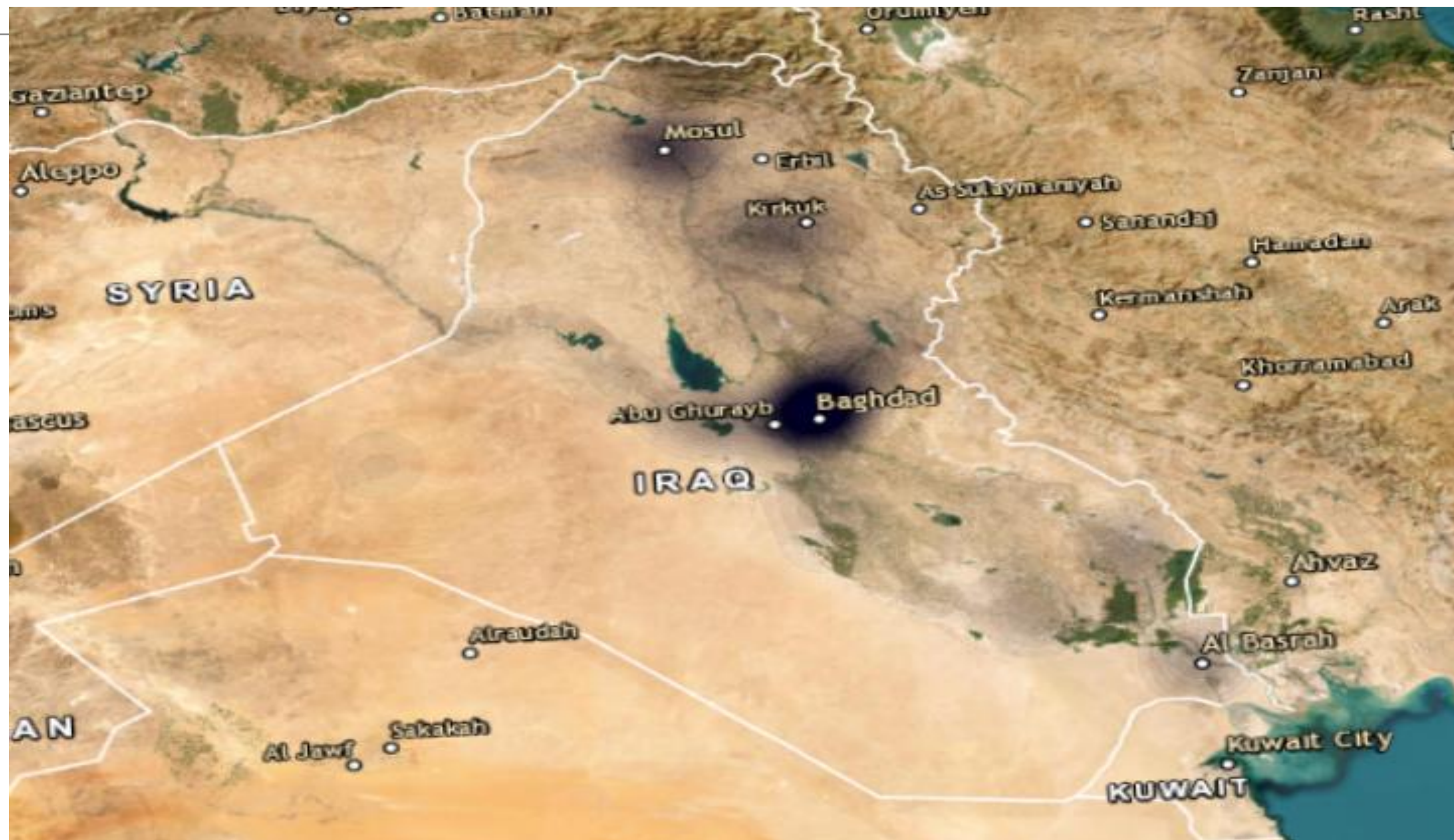


Violence against civilians' graphic



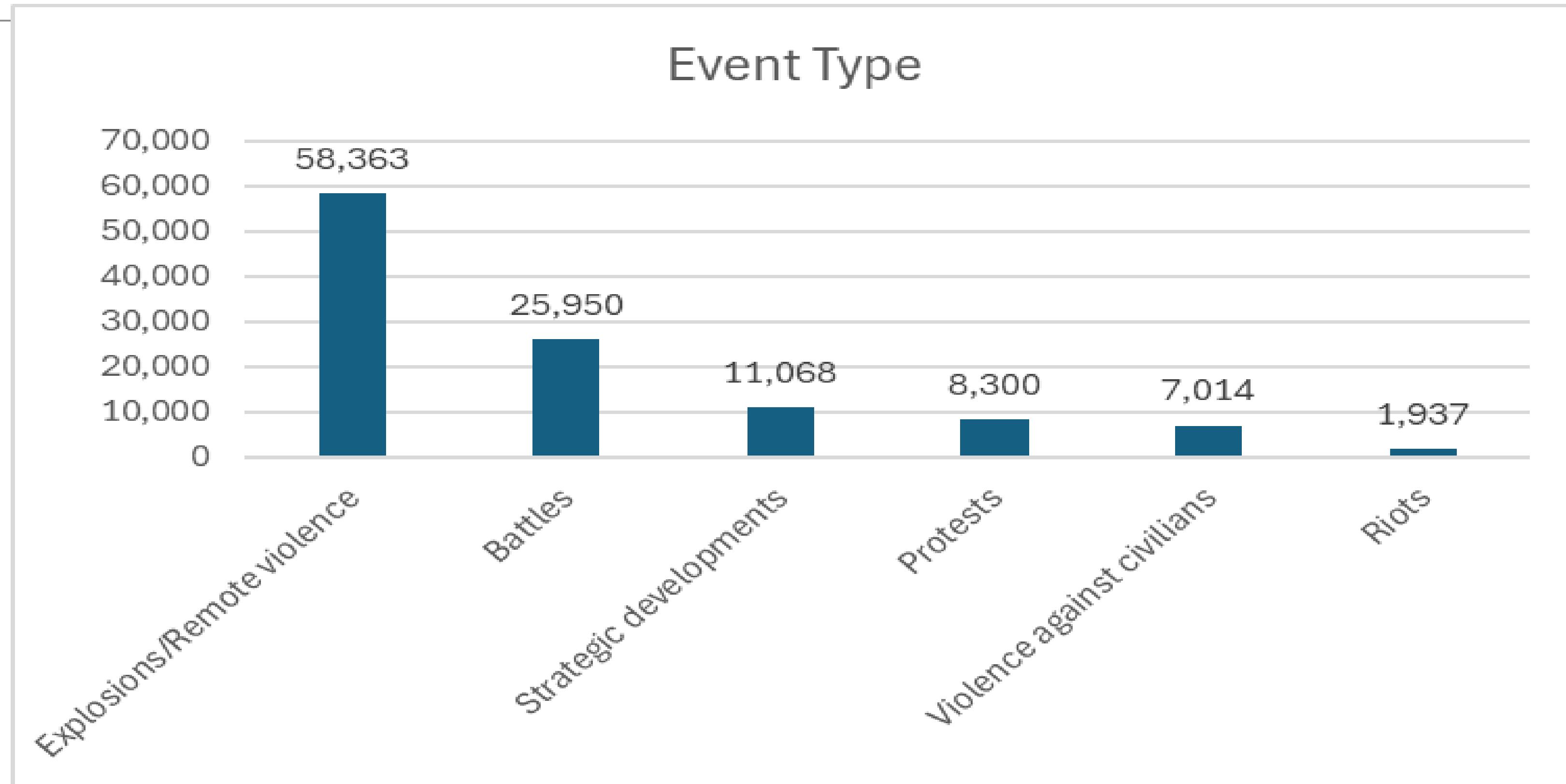


How was the event type distributed?





How were the event type distributed graphic

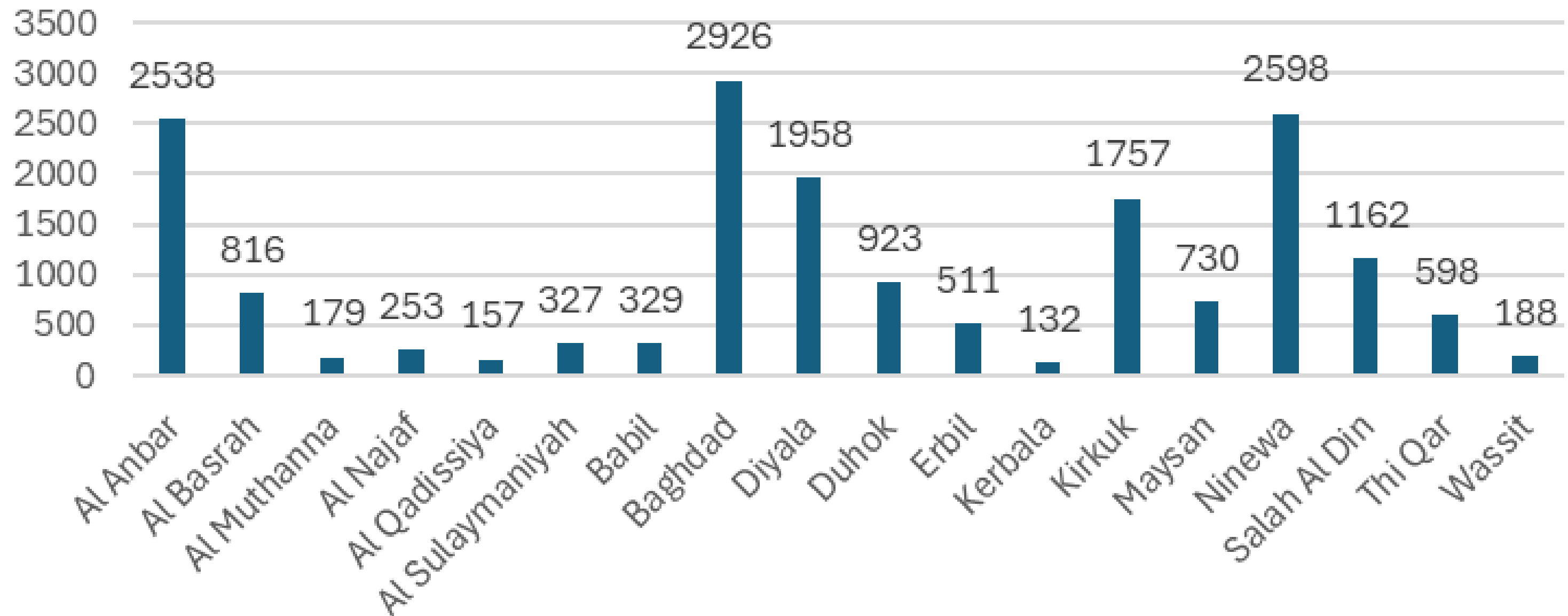




Distribution of event types graphic

Sum of Count

Distribution of Event Types



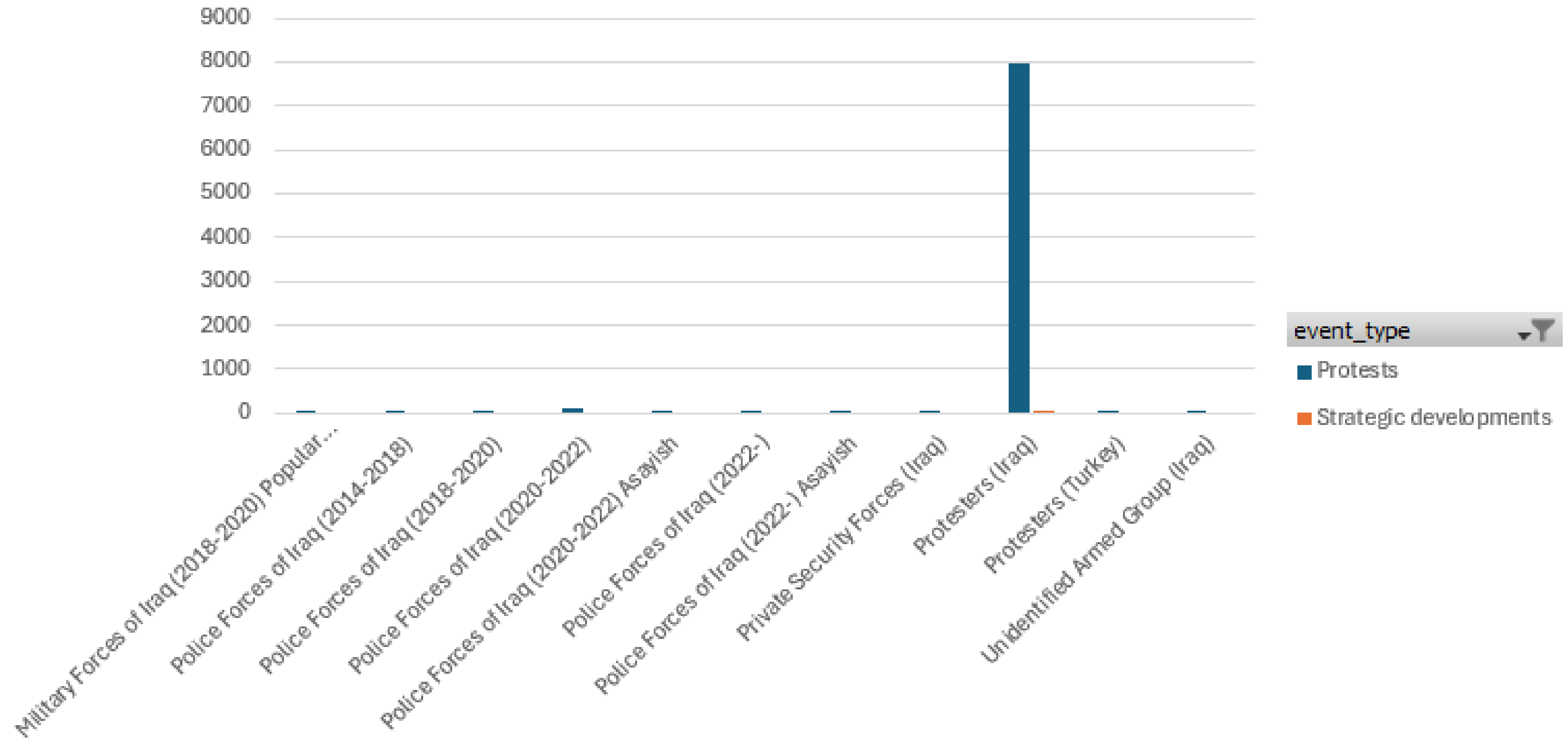
admin1 ▼



Who were the actors in each event type?

Sum of Count

Top 10 Actors in Each Event



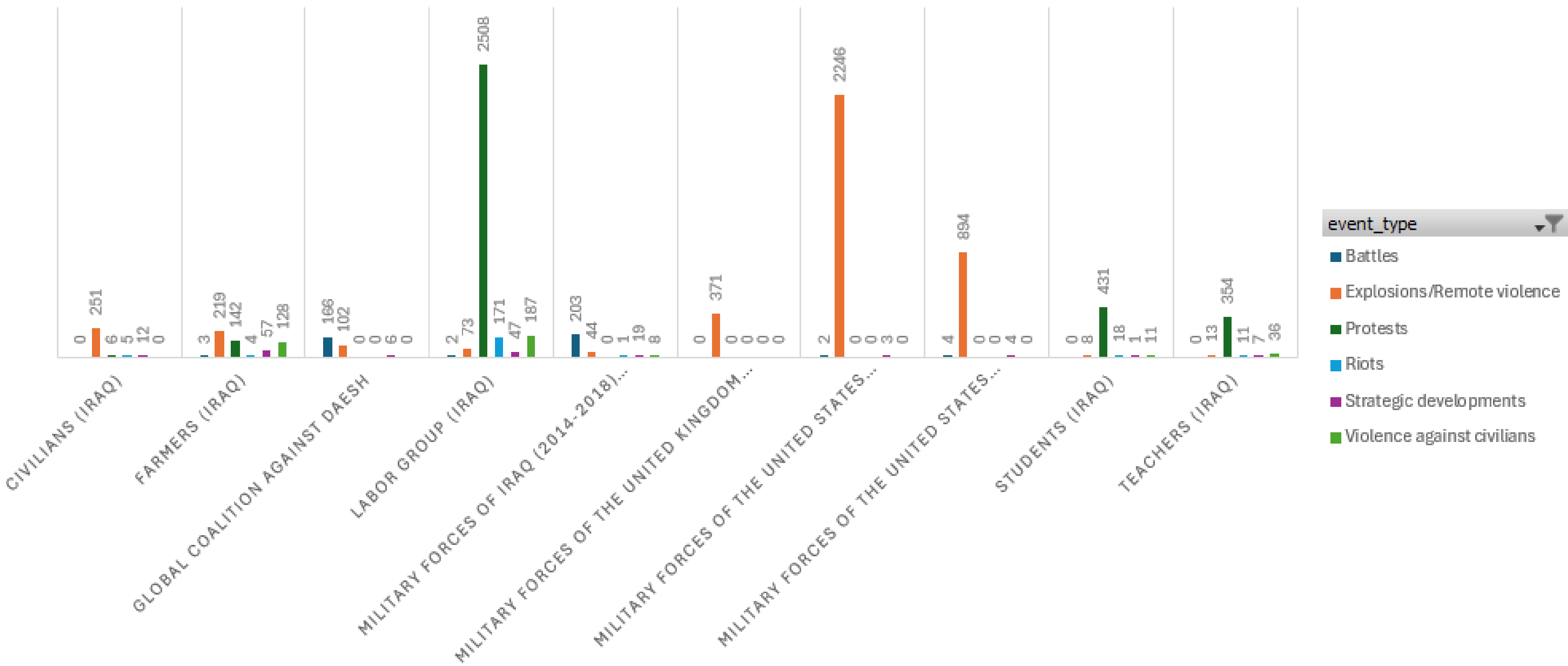
actor 1



What were the main associated actors in each event type?

Sum of Count

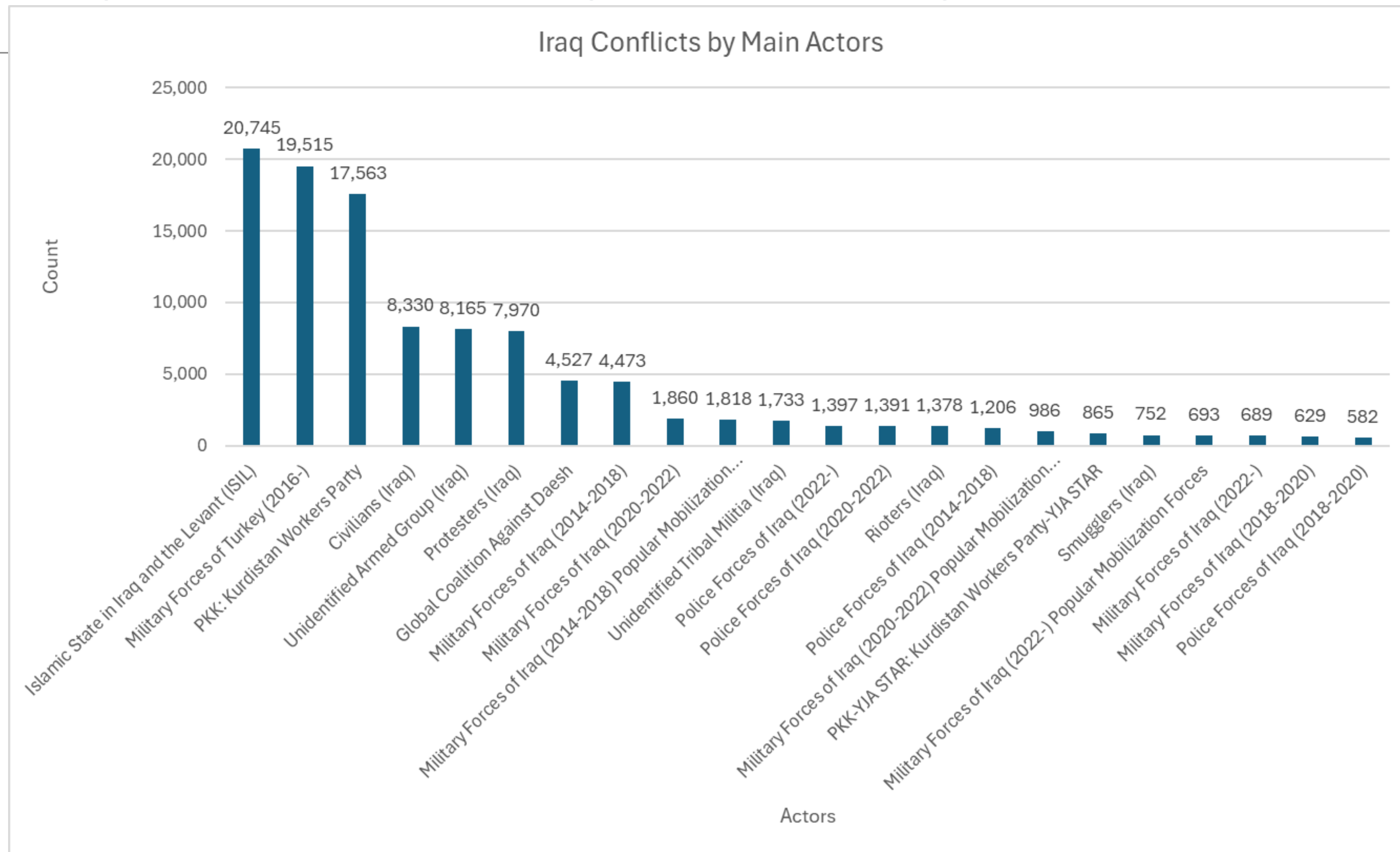
Top 10 Associated Actors by Event Types



assoc_actor_1

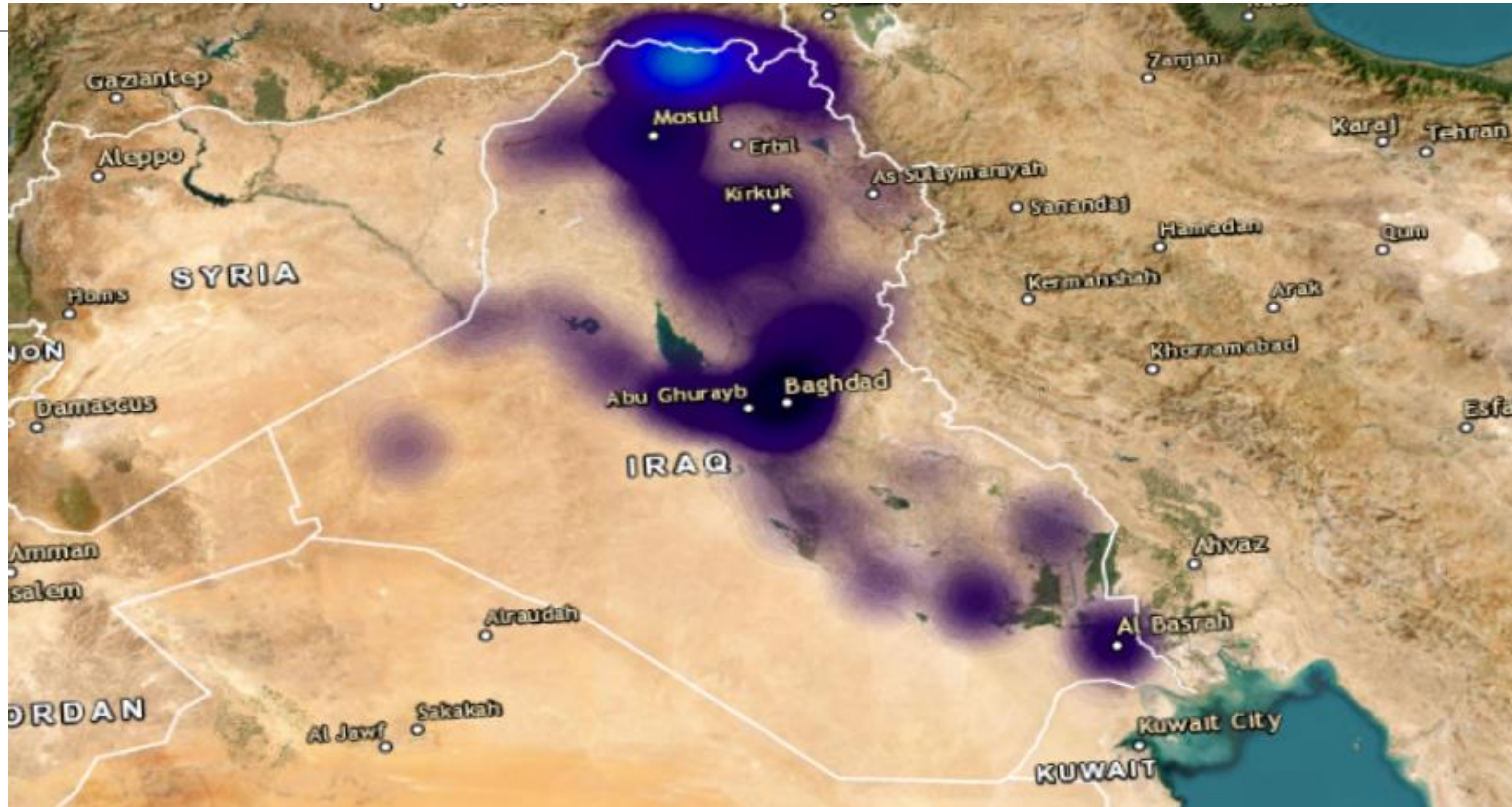


Who were the main actors in interactions?





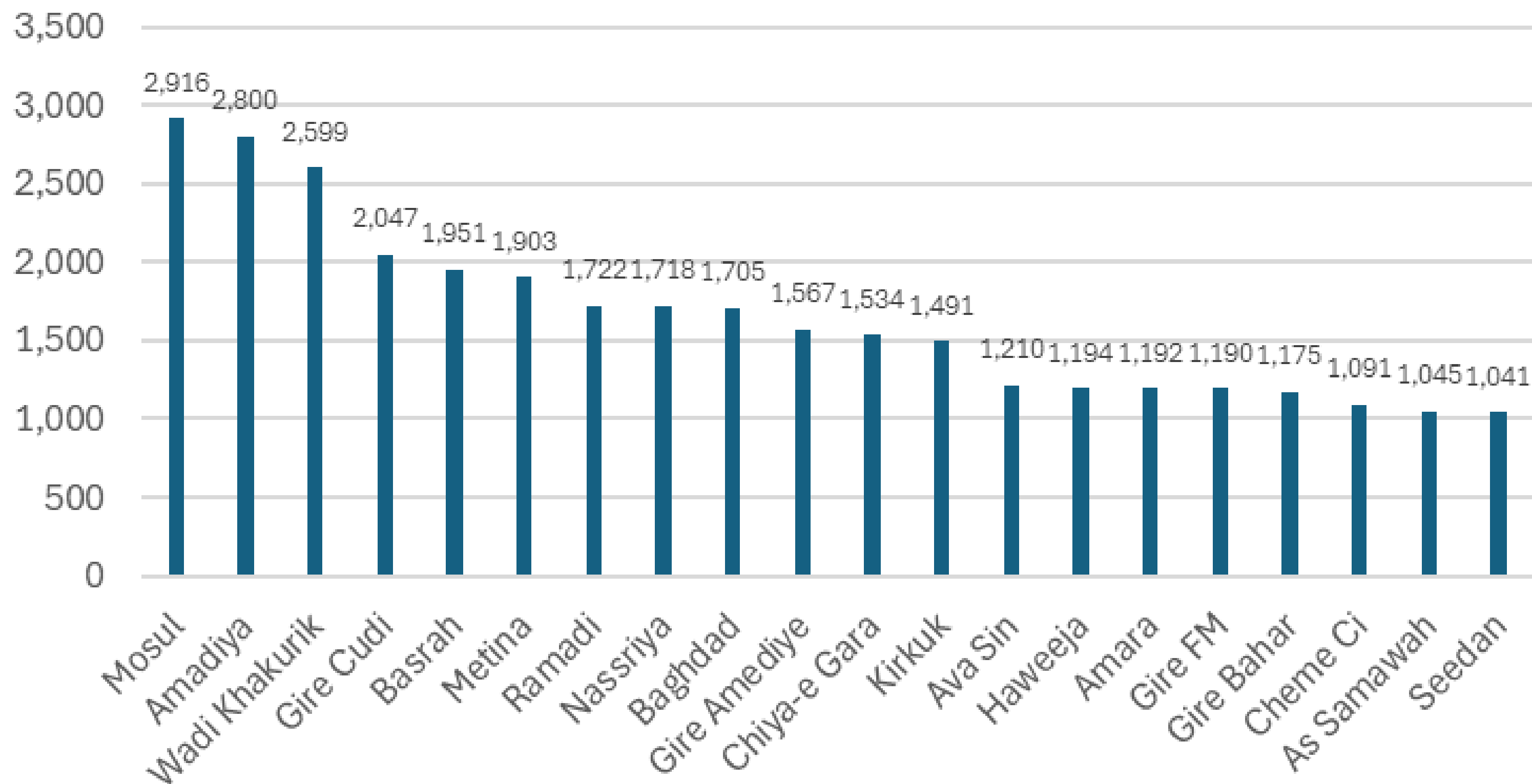
Show me distribution by location?





Where are the conflicts concentrated?

Top 20 Locations by Conflict



Thank You!

**Xavier Reyda, Thomas
Wickham, James Gotch,
Sabastian Lemansky**